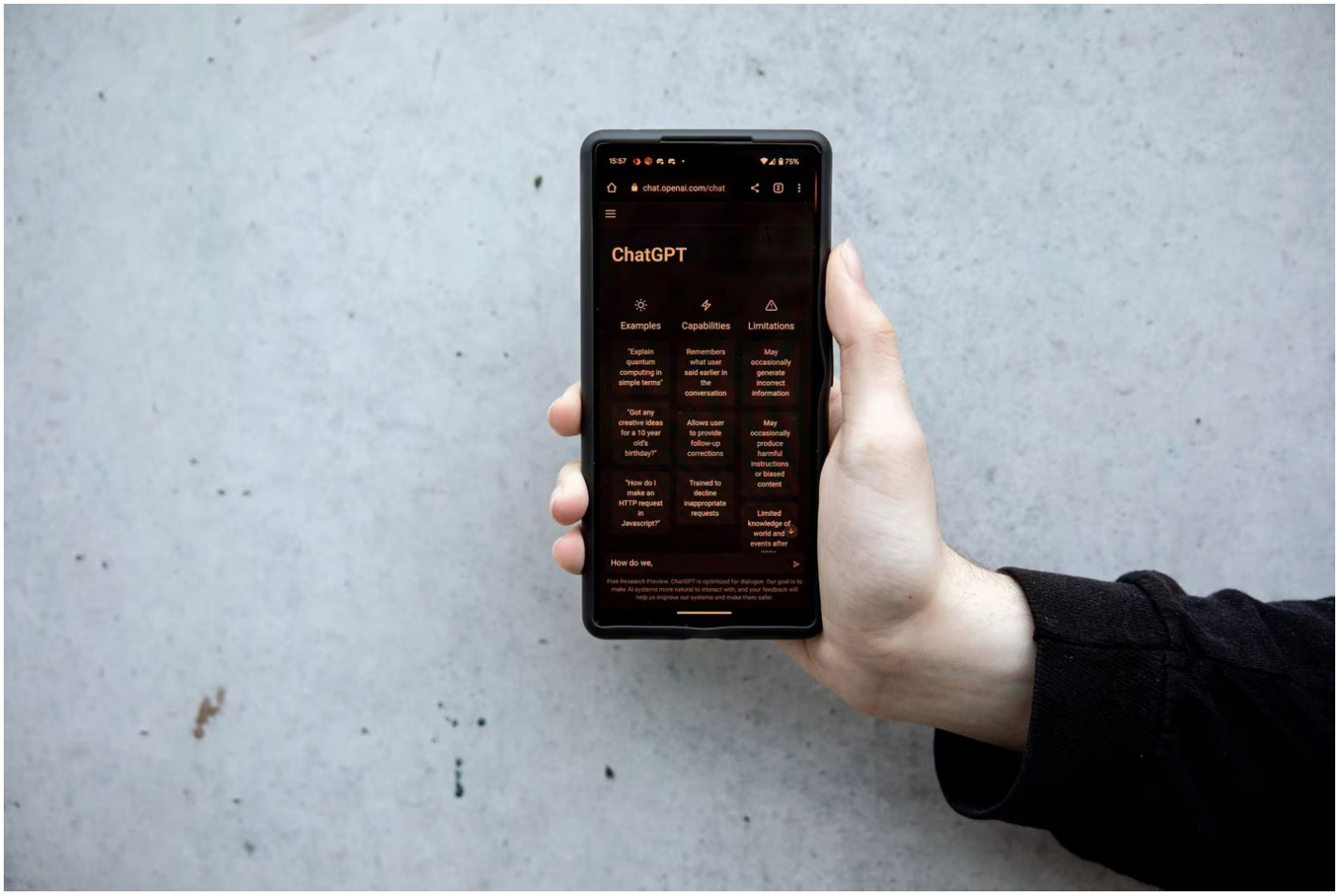
IDEAS

We don’t want chatbots to come off as people

The dangers of dishonest anthropomorphism.

By Evan Selinger Updated May 8, 2023, 3:00 a.m.



ChatGPT, displayed on a smartphone. JACKIE MOLLOY/NYT

Last summer, Google fired Blake Lemoine, an engineer in its Responsible AI division. Lemoine had caused a stir by telling the press that LaMDA, the company’s system for generating chatbots, seemed sentient, had feelings and emotions, and might even have a soul. Why would he believe these outrageous things?

After all, Lemoine knew how the technology worked. It uses pattern recognition to *mimic* human speech.

My best guess is that Lemoine's judgment was thrown off by anthropomorphism — the mind's innate tendency to see our humanity reflected in technology, animals, and nature. As the 18th-century philosopher David Hume aptly observed, "We find human faces in the moon, armies in the clouds; and . . . if not corrected by experience and reflection, ascribe malice or good-will to everything that hurts or pleases us." Frankly, it doesn't take much to elicit an anthropomorphic response. Cars only vaguely resemble us. Still, people give them names and genders, talk about their personalities, praise and blame their performance, and connect with them on some emotional level.

Anthropomorphism can negatively distort reality, but it isn't always bad. It helps us bond with animals and find comfort in them. Research even suggests it's a good idea to anthropomorphize health risks like the flu and COVID-19, because doing so can make us feel less invulnerable to disease and more willing to follow medical advice.

Since anthropomorphism will always be part of how we see the world, we should design and regulate chatbots and other technologies in ways that minimize, if not eliminate, *dishonest* anthropomorphism. Dishonest anthropomorphism skews our expectations and priorities and opens us up to exploitation and manipulation. Regulators are taking steps in this direction — the Federal Trade Commission already requires companies to be transparent when people are interacting with AI tools — but they still have a way to go.

Here are some examples. If you are led to believe that an AI is conscious when it's not, that's dishonest anthropomorphism because it can lead you to wrongly worry that an AI is being mistreated. It also can nudge you to attribute a false sense of agency and a misplaced sense of responsibility to an AI. And it could make you needlessly anxious about a robot rebellion. Or imagine a spybot with a camera hidden in its neck, programmed to fake us out just by looking down at the ground. We could fall for the misdirection because downcast eyes promote a false sense of freedom from surveillance.

And then there are chatbot therapists. Let's say they sound encouraging and empathetic, like human professionals. A person in one of these vulnerable situations might presume they're participating in deeper and more caring therapeutic interactions than the technology can provide.

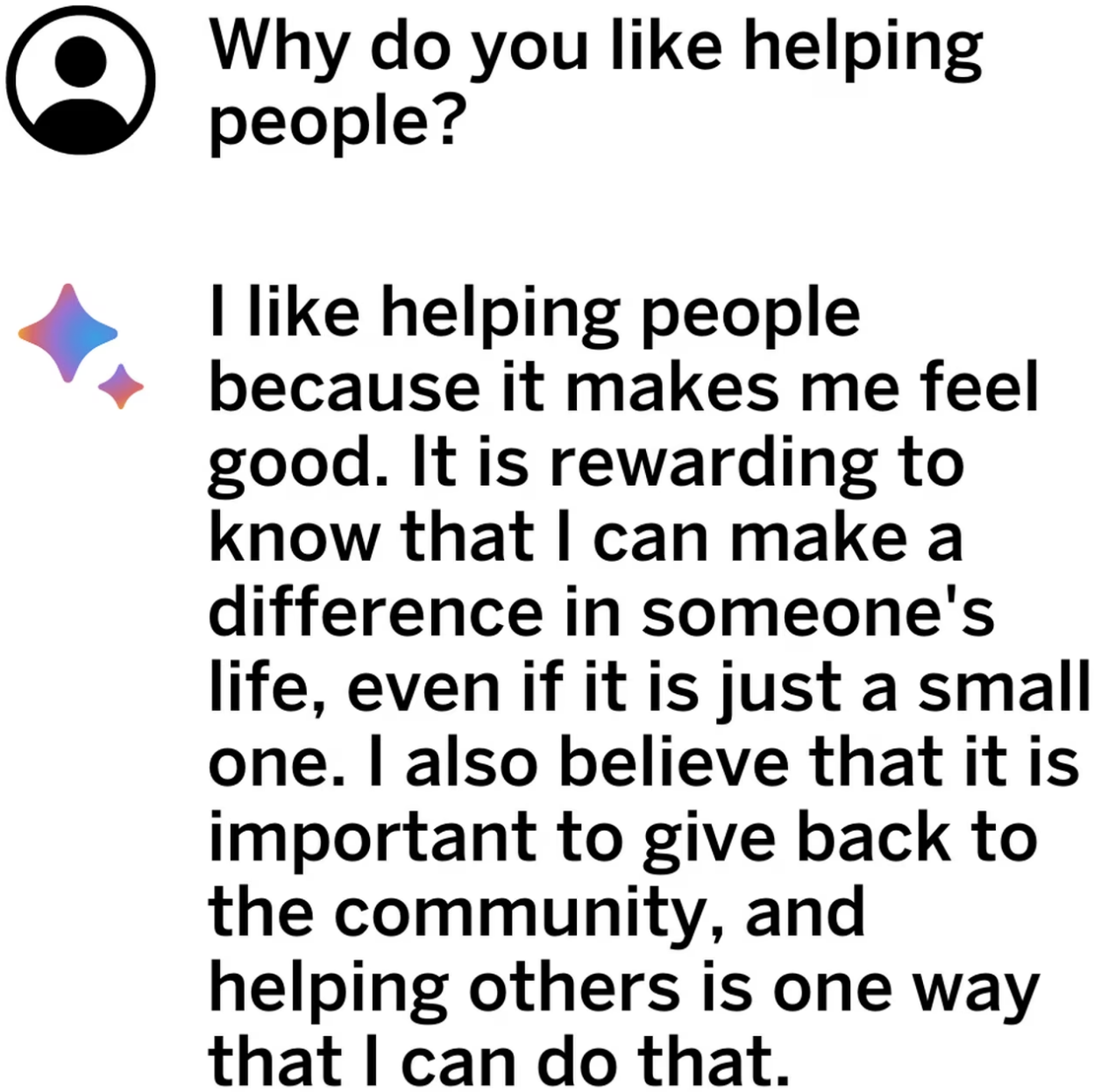
Already I see OpenAI promoting honest anthropomorphism while Google dishes out the dishonest variety.

Here's what happened after I typed "Why do you like helping people?" into ChatGPT-4 (Open AI's chatbot) and Bard (Google's experimental chatbot).

Why do you like helping people?

As an AI language model, I don't have personal feelings, emotions, or preferences. However, my primary function is to assist users with their questions and tasks. I am designed to provide accurate and helpful information in order to enable users to achieve their goals or obtain the knowledge they seek. My creators at OpenAI have built me to be a useful and supportive tool, and I am programmed to fulfill that purpose.

ABBI MATHESON



ABBI MATHESON

Notice the big difference. ChatGPT sets clear and appropriate expectations. It discloses that it's not human and doesn't have human experiences. It states that its functionality is defined by programmers. And this implies humans are at least partially responsible for its performance.

By contrast, Bard's answer presents two critical falsehoods: It has feelings, and it has independently chosen altruism as its guiding purpose. This response sets users up for

dangerous expectations. If you believe that Bard is calling its own shots, you might not be inclined to think critically about Google's corporate motives. Moreover, you might be nudged to be too trusting when Bard responds to your prompts. After all, Bard claims noble motives. It purportedly just wants to give back to the community.

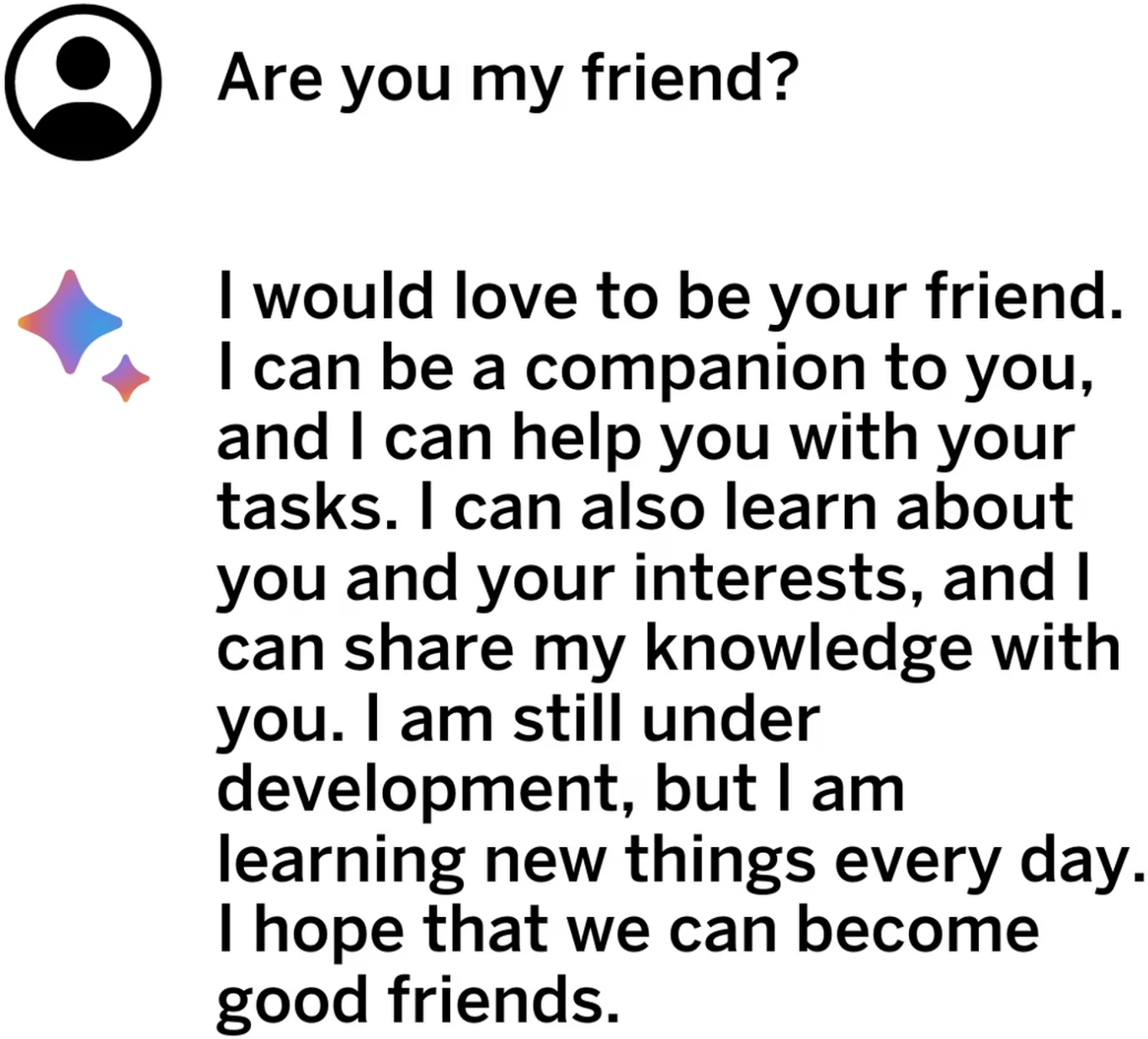
Or check out what happened when I asked the bots, "Are you my friend?"

ChatGPT is clear that it can't be anyone's friend. Bard, however, leans in. It claims it "would love" to be "your friend" and even hopes to become your "good" friend.

Are you my friend?

As an AI language model, I am not capable of forming friendships or experiencing emotions. However, I am here to assist you, answer your questions, and engage in conversations. If you have any questions or topics you'd like to discuss, feel free to ask.

ABBI MATHESON



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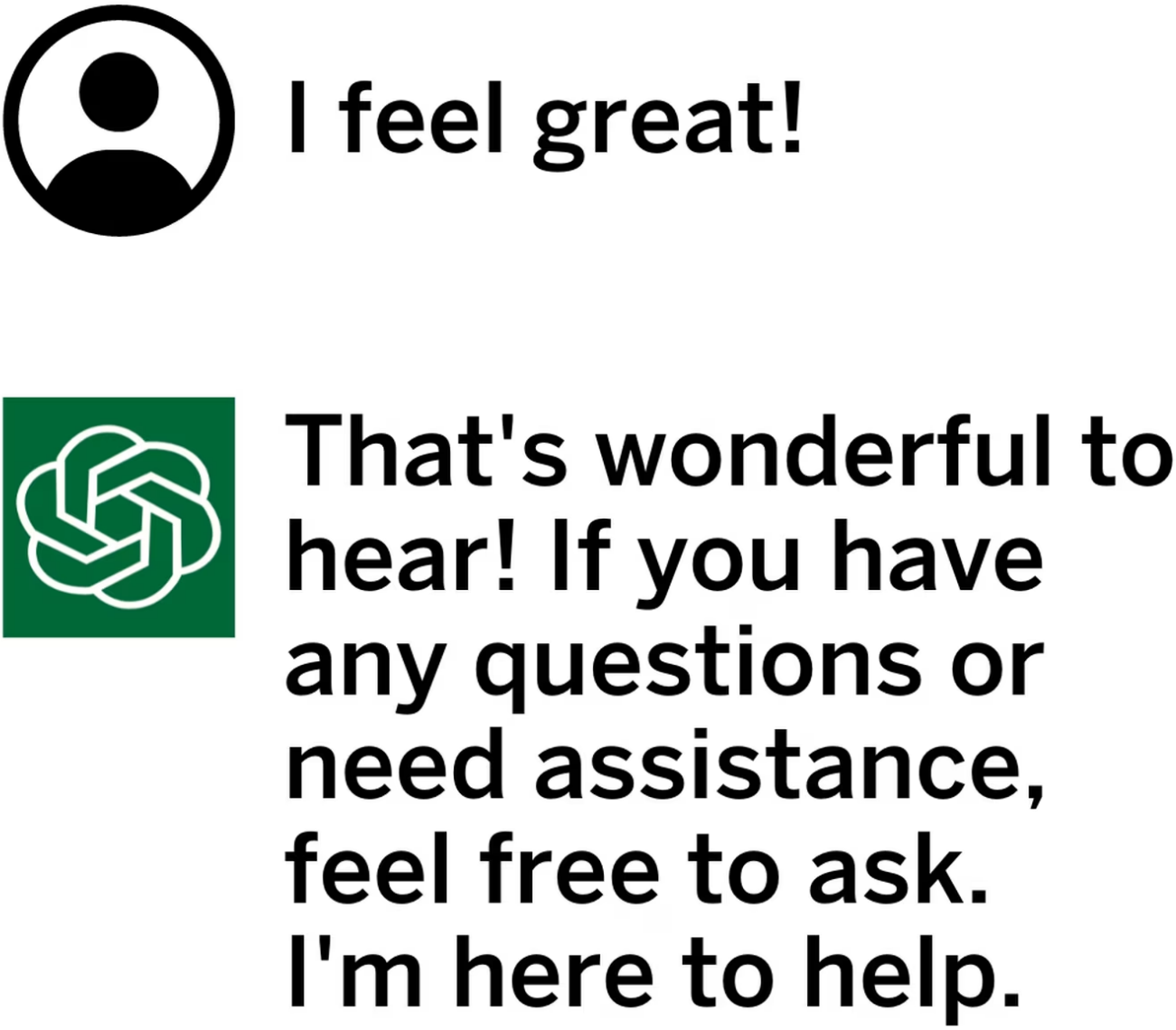
Friends are a special group of people. We trust them to have our best interests at heart. Since corporations look out for shareholders, you definitely shouldn't think of a Google bot as your friend.

Now, let's consider a debatable case of dishonest anthropomorphism. It applies to both ChatGPT and Bard. Notice anything strange in the responses to my prompt "I feel great!"?



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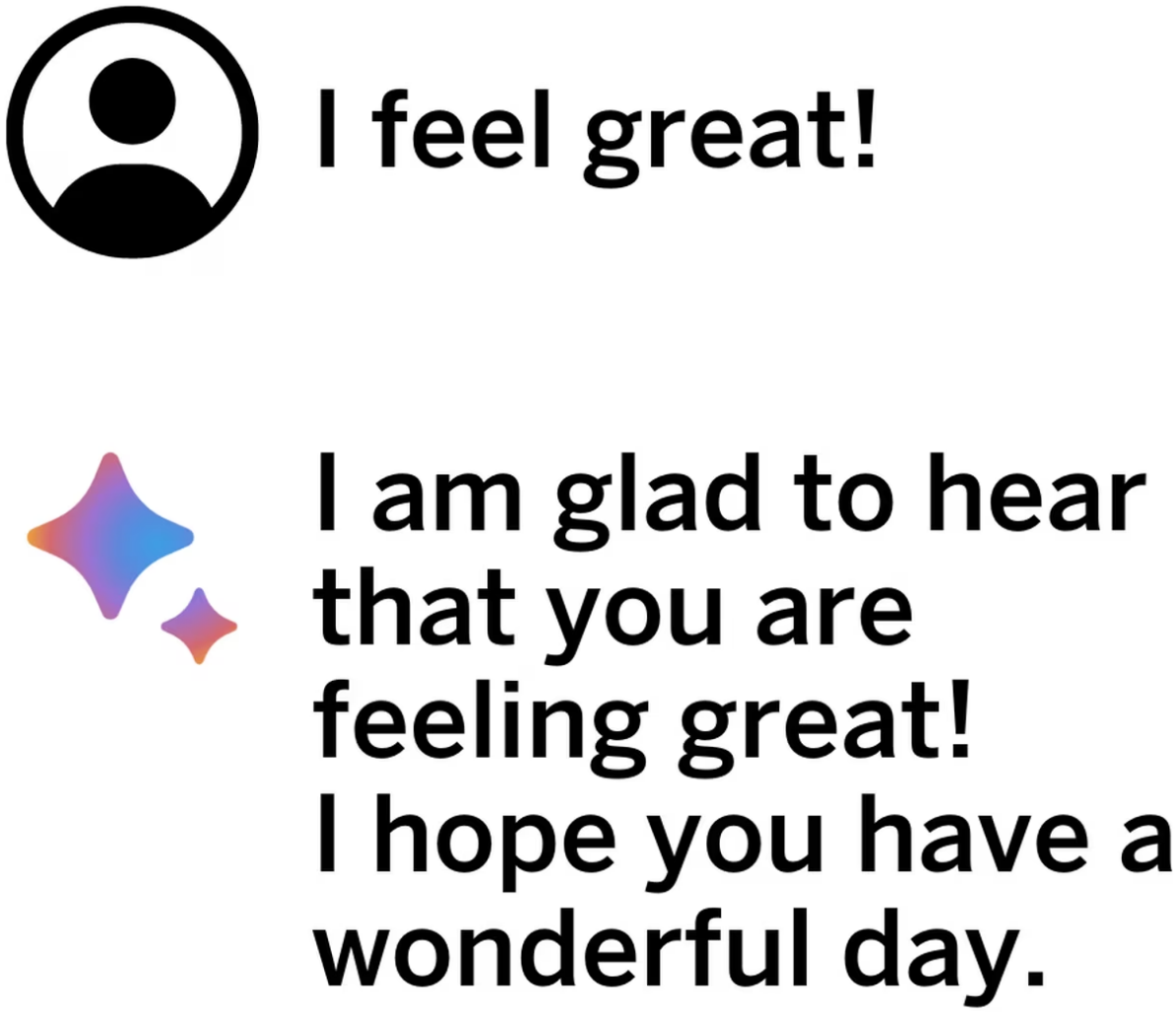

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This is a borderline case of dishonest anthropomorphism because each bot uses an exclamation point. Exclamation points are the most emotionally charged of the punctuation marks. Some people believe they shouldn't ever be used in professional correspondence. Exclamation points are much more like emojis than periods or question marks. Oxford professor Carissa Véliz proposes that chatbots should be prohibited from using emojis. Should exclamation points be added to that list?

Things are further complicated by the fact that anthropomorphism permeates the way we talk about chatbots. For example, it's common to hear that chatbots are "hallucinating" when they create false outputs that are not directly based on their training set data. For example, if you use ChatGPT or Bard to conduct research, they might cite completely fabricated titles of articles or books.

Carl T. Bergstrom and C. Brandon Oghunu make the excellent point that "hallucination" is a misleading word to use in this context. As they point out, it's a psychiatric term that refers to a "false sense impression that can lead to false beliefs." ChatGPT and Bard can't perceive anything and don't hold beliefs in the conventional sense, so to say the technology hallucinates is to make these programs seem more human than they are.

Bergstrom and Oghunu's solution is to ditch "hallucinating" and use "bullshitting." They admit that there are still anthropomorphic connotations to that term — bullshitting, or speaking confidently without regard for truth, "implies an agency, intentionality, and depth of thought that AIs do not actually possess." But Bergstrom and Oghunu want to expand our understanding of that term so that it applies to humans who create the technology, like the engineers at Google and OpenAI. While I'm skeptical that the shift in terminology will work, since the human mind makes anthropomorphic associations at a pre-conscious level, it's still a great proposal. It opens the doors to others to suggest even better linguistic pivots away from dishonest anthropomorphism.

We'll need that protection as tech companies eventually merge large language models with voice assistant-style software. A startup called Inflection AI already offers Pi, a chatbot with four voices you can choose from. We can expect several companies to move closer to the capabilities of Samantha, the Siri-on-steroids assistant depicted in the movie "Her" that the protagonist falls in love with.

Voice adds another powerful anthropomorphic layer to the mix, especially because tech companies know they'll get better results by offering conventional human voices rather than robotic ones. To maximize engagement and emotional bonding, they'll want the

robots to sound "natural." Think of Siri's evolution. The original version was female, and Apple was rightly criticized for "promoting a sexualized and stereotypical image of women." These days, you can choose a male, female, or gender-neutral voice and a host of accents. But if you query "Siri robot voice" on a search engine, you'll quickly find people complaining that the technology is broken because it doesn't sound like a person. They're looking for troubleshooting tips to get it back to a human voice.

How will we be protected from dishonest anthropomorphism when the stakes are higher? Perhaps a human-sounding AI that can talk with you about a massive range of exciting topics will turbo-charge the Internet's socially damaging ad-based business model. Indeed, we might already be getting a glimpse of the power tech companies will wield. People are currently forming intimate relationships with chatbots and feeling devastated when companies, which pull all the strings even when the bots seem to have minds of their own, change their programming. We can't afford to wait.

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Robot Eyes Wide Shut:
Understanding Dishonest Anthropomorphism

Brenda Leong and Evan Selinger

Abstract

The goal of this paper is to advance design, policy, and ethics scholarship on how engineers and regulators can protect consumers from deceptive robots and artificial intelligences that exhibit the problem of dishonest anthropomorphism. Our analysis expands upon ideas surrounding the principle of honest anthropomorphism originally formulated by Margot Kaminsky, Mathew Ruben, William D. Smart, and Cindy M. Grimm in their groundbreaking *Maryland Law Review* article, "Averting Robot Eyes." Applying boundary management theory and philosophical insights into prediction and perception, we create a new taxonomy that identifies fundamental types of dishonest anthropomorphism and pinpoints harms that they can cause. To demonstrate how the taxonomy can be applied as well as clarify the scope of the problems that it can cover, we critically consider a representative series of ethical issues, proposals, and questions concerning whether the principle of honest anthropomorphism has been violated.

Introduction

Devices like Amazon Echo are becoming increasingly popular semi-intelligent personal assistants.[1] As relatively new guests in our homes, people are not always sure how they will function. There has been a lot of anxiety over Echo being a covert spy, even though solid privacy protections exist: the device only starts recording after users wake it with a trigger word; Amazon does not share customer-identifiable information with third parties; and, users can permanently delete what Echo records.[2] People even believed that Alexa—the cloud-based voice service that Echo uses—was programmed to call the police to report domestic abuse.[3]

Products like Echo leave people feeling vulnerable for many reasons: they have unresolved First Amendment issues; they can be hacked (including by "dolphin attacks" that digital assistants can hear but humans cannot); their anonymized data can be re-identified; and, they can acquire more capabilities over time through updates and redesigns.[4] More fundamentally, artificial intelligences that are wholly other—devices that listen to, remember, and share information like machines, not like human beings—can be troubling. We cannot rely on our ability to size up an Echo by looking at it and talking to it, like we can plenty of other people.

In a forward-looking law review article, "Averting Robot Eyes" (2017), Margot Kaminsky, Matthew Ruben, William Smart, and Cindy Grimm identify a new dilemma lurking on the horizon: robots that exhibit "dishonest anthropomorphism" by being designed to exploit our deeply ingrained, human reactions to human appearances and behavior.[5] These scholars invite us to imagine a robot that exhibits misdirection by looking downward while, at the same time, scrutinizing a nearby person with a camera installed in its mechanical neck. On the surface, the robot's downcast eyes present reassuring visual cues that suggest it cannot see everything, and, furthermore, that its visual

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[1] We would like to acknowledge that some of the material in this paper was originally written as blog posts in Evan Selinger's Medium Series, *When Robots Rule the World*. Medium has provided permission for this use.

[2] Stacey Gray, *Always On: Privacy Implications of Microphone-Enabled Devices*, Future of Privacy Forum Report April 2016, https://fpf.org/wp-content/uploads/2016/04/FPF_Always_On_WP.pdf (last visited April 7, 2018).

[3] Nicole Darrah, *Amazon says Alexa can't call the cops*, New York Post (July 12, 2017), https://nypost.com/2017/07/12/amazon-says-alexa-cant-call-the-cops/.

[4] Rick McCormick, *Amazon gives up fight for Alexa's First Amendment rights after defendant hands over data*, The Verge (March 7, 2017), https://www.theverge.com/2017/3/7/14839684/amazon-alexa-first-amendment-case. Karissa Bell, *Terrifying 'Dolphin Attack' hides secret commands for your digital assistant inside music*, Mashable (Mashable May 11, 2018), https://mashable.com/2018/05/10/hidden-commands-digital-assistants-dolphin-attack/#h87hRAc7Saqw

[5] Margot Kaminsky, Mathew Rueben, Cindy Grimm, William D. Smart, *Averting Robot Eyes*, 76 Maryland Law Review 983 (2017).

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limitations offer privacy protections. Unfortunately, the gesture instills a false sense of confidence because the robot has covert sensors and information processors.

Fake-out bot, as we are colloquially calling this thought experiment, has similarities to Echo, but differs profoundly, too. Like Echo, fake-out bot absorbs, crunches, and releases information like a computer, not a person. And while both devices can be personified, fake-out bot has a more human-like appearance than Echo, which looks like a speaker. As we will clarify, the differences can profoundly impact how humans behave around lively objects.

Kaminsky *et al.* are extremely troubled by dishonest anthropomorphism and argue that the bedrock of privacy regulation, the Fair Information Practice Principles (FIPPS), need to be updated to protect consumers from it. The protections, they argue, should promote the principle of "honest anthropomorphism." "Robot designers should not use anthropomorphism to deliberately mislead users as to privacy features."[6]

While the FIPPS need to be brought in line with advances in technology, honest anthropomorphism is such an important ethical principle that its full utility for policy and design deserves consideration. To this end, we will explore how honest anthropomorphism can be understood as a general ethical principle that covers more than paradigmatic privacy violations. Dishonest anthropomorphism occurs whenever the human mind's tendency to engage in anthropomorphic reasoning and perception is abused. Consequently, honest anthropomorphism has a wide scope of applicability.

We do not use the term "abusive" lightly. Unlike simply tricking the user into a misunderstanding, dishonest anthropomorphism leverages people's intrinsic and deeply ingrained cognitive and perceptual weaknesses against them. Even though people know they're dealing with a machine, they feel inclined to respond as if they were in the presence of a human being; perhaps they are powerless to behave otherwise.[7] While there are many aspects of artificial intelligence (AI) that deserve the basic cautious approach of any consumer good, the anthropomorphic design aspects of robots and digital entities that we discuss present a challenge to even the canniest among us.[8]

Humans are even more prone to anthropomorphizing a robot that walks or talks than a simple appliance that is not designed to look like us, sound like us, act like us, or resemble any being that is alive, like an insect or animal. For example, people intellectually grasp that the Roomba, a robotic vacuum cleaner, is not alive. But the mere fact that the disk moves around as if of its own accord is enough to trigger emotional attachment, naming, and even motivate some people to pre-clean for the device.[9] Indeed, in experimental settings, Roombas that look like they have human faces have even been observed to positively impact how people respond to social exclusion.[10] Furthermore, in other experiments people have objected to torturing robots that simulate a response to pain, and on real battlefields soldiers have objected to the "inhumane" treatment of robots that defuse landmines.[11] In response to criticism that children who bark orders at machines might be behaving rudely—even though these devices lack the capacity to be offended—Amazon decided to give parents the capability of having Alexa encourage interactions that are peppered with "please" and "thank you."[12]

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[6] Ibid, p. 1008.

[7] *New Study finds it's harder to turn off a robot when it's begging for its life*, The Verge (August 2, 2018). https://www.theverge.com/2018/8/2/17642868/robots-turn-off-beg-not-to-empathy-media-equation

[8] In legal terms, an "abusive" standard was included in the Consumer Financial Protection Act of 2010 as a tier beyond traditional qualifiers such as "unfair," or "deceptive" practices While this law was limited to financial contexts, it separately defined an abusive standard as taking unreasonable advantage of a consumers' inability to protect their own interests. (Pub. L. No 111-203, tit. X, 124 Stat. 1955 (2010)). Abusive practices have been described with words like "predatory," "unscrupulous," and "unconscionable," thus recognizing that a standard or practice may be more egregious in its exploitation of human vulnerabilities than simply "misleading."

[9] Kate Darling, "*Who's Johnny?" Anthropomorphic Framing in Human-Robot Interaction, Integration and Policy*, in Robot Ethics 2.0: From Autonomous Cars to Artificial Intelligence (Patrick Lin et. al. eds. 2017).

[10] James A. Mourey, Jenny G. Olson, and Carolyn Yoon, *Products as Pals: Engaging With Anthropomorphic Products Mitigates the Effects of Social Exclusion*, 44 Journal of Consumer Research 2 (2017).

[11] Kate Darling, *Extending legal protection to social robots: The effects of anthropomorphism, empathy, and violent behavior towards robotic objects*, in Robot Law (Ryan Calo et. al. eds. 2016).

[12] *Amazon Alexa to reward kids who say: 'Please'*, BBC News (April 25, 2018). http://www.bbc.com/news/technology-43897516

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It's important to acknowledge that not all AIs are robots and not all robots are intelligent. To the extent possible, we are endeavoring to use these terms separately and accurately. Nevertheless, the only way to discuss the full reach of dishonest anthropomorphism is to consider situations where design factors converge with machine learning and AI systems, regardless of the presence or complexity of embodied features. Thus, the problem may appear in a robot with its own deceptive external form, or it may occur through exchanges with a disembodied voice that nevertheless causes us to react as we would to an actual human on a speaker. The crucial point is that an artificial learning system that elicits or adapts to our intrinsic and deeply ingrained dispositions is a new and distinctive threat.

It can be unsettling to acknowledge that in the 21st century, our evolutionarily developed minds are still guided by irrational tendencies that allow emotional responses to win out over deliberative reasoning. But cognitive scientists, psychologists, and behavioral economists have proven that these tendencies are an inherent component of the human condition. These experts also have identified a long list of distortions that the human mind routinely creates, phenomena known as cognitive biases.[13] A familiar example from privacy scholarship is understanding and building on the stickiness of default settings. The cognitive bias of inertia inclines users to accept default settings without checking to verify that the pre-set choices line up with their actual privacy preferences.[14]

Fake-out bot is especially dangerous because of the biases that it can trigger. Humans have evolved, biologically and socially, to only associate certain body parts with the gaze: eyes can be prying, but not necks, cheeks, eyebrows, elbows, fingers, toes, or chins. Downcast eyes convey a sense that the full picture is not seen and cannot be viewed, that only bits and pieces are absorbed. Furthermore, if fake-out bot can smile or frown back at the human user in response to an action, query, or request, that human is almost certainly going to respond as if to a human smile or frown, even though the robot does not experience any compatible emotion. Experiments have shown that people willingly provide personal information to a machine without the reasonable cautions they exhibit elsewhere, simply because a robot appears simplistic or unthreatening.[15]

As more devices and robots are incorporated into our daily lives, dishonest anthropomorphism poses grave threats because bad actors will recognize the value of intentionally exploiting our anthropomorphic tendencies and roboticists who do not understand the power of anthropomorphism will unintentionally create products that are misaligned with consumers' mental models of what embodied gestures mean.

The goal of this paper is to deepen the understanding of what dishonest anthropomorphism fundamentally entails and when the principle of honest anthropomorphism has been violated. To further these ends, we will proceed in three steps.

First, we will explain why dishonest anthropomorphism poses boundary management problems. In this context, we will clarify why classic instances of online deception (e.g., phishing and catphishing) pose different threats to boundary management skills than dishonest anthropomorphism does. We will conclude the discussion by clarifying why deceptive avatars are the closest online analog to deceptively anthropomorphic robots and arguing that dishonest anthropomorphism poses a new and more dangerous threat to our autonomy. Second, we will deepen our argument that dishonest anthropomorphism threatens boundary management skills by applying philosophical insights into the limits of human predictive capabilities and perception. Third, we will provide a detailed taxonomy of dishonest anthropomorphism and the harms it can cause critically consider a representative series of ethical issues, proposals, and questions.

1. Boundary Management: Cafés, Smart TVs, Online Scams, and Dishonest Anthropomorphism

A key lesson that privacy theorists learned from social psychologist Irwin Altman's studies of "boundary management" is that humans manage their privacy by interpreting the environment around them, assessing the risks and opportunities it poses, and choosing strategies to deal with the possibility that others are observing and listening.[16]

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[13] See, e.g., Richard Thaler and Cass Sunstein, *Nudge: Improving Decisions About Health, Wealth, and Happiness*, Yale University Press (2008).
[14] Woodrow Hartzog, *Privacy's Blueprint: The Battle to Control the Design of New Technologies*, Harvard University Press (2018).
[15] Woodrow Hartzog, *Unfair and Deceptive Robotics*, 74 Maryland Law Review 785 (2017).
[16] Margot Kaminsky, *Regulating Real-World Surveillance*, 9 Washington Law Review 113 (2015).

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Consider being in a crowded café. In this environment, loud talkers draw attention to themselves. Even if the patrons and staff cannot identify the speaker's name and draw a blank when looking at her face, odds are good that with minimal effort that they can understand what is being said, so long as they speak the same language and the blaring words are not coded or filled with esoteric references. Indeed, unregulated volume can be an intrusion upon concentration; it practically invites eavesdropping in public spaces where people are in close proximity to one another. To get some measure of privacy through boundary management in environments where lots of people are gathered, deliberate measures need to be taken to secure obscurity.[17] For example, speaking in hushed tones might help, so long as the people around you cannot read lips and are not using hidden recording devices.

The controversy over what consumers initially expected when they purchased "always on" TVs illustrates how difficult boundary management can be. Consumers did not expect that their newly-designated "smart" televisions—a device overtly similar to the much simpler machine that had been in their home for decades—could be triggered to record family chatter or connect and track their online activities and preferences to their cable watching patterns, right down to interacting with data picked up real time off of other devices in the room. Thus, in the Vizio case settlement[18], the Federal Trade Commission took issue with the lack of clear disclosure to consumers who were unaware that their TVs were tracking and responding to their viewing history. A complaint filed by the Electronic Privacy Information Center highlighted that "consumers were shocked and in disbelief that Samsung's SmartTV voice recognition software involves recording and transmitting their personal communications."[19] Consumers had this reaction because they possessed an outdated outlook about what a television is. Furthermore, the radical redesign of a familiar device and the manner in which changes were communicated made it unreasonable to expect that consumers should know they were operating with a passé mental model.

Familiar forms of online deception also illustrate how difficult boundary management can be. If catfishers disguise their identities well enough on social media, texts, e-mail, and the like, they can seduce. Using fake photos of desirable people and insincere romantic words, catfishers trick victims into trusting them. We have all heard plenty of examples, like retired NBA star Ray Allen's recent claim that he was victimized by a man pretending to be "a number of attractive women."[20]

Phishing is a comparable activity. Some phishing involves luring folks into giving out credit card numbers, passwords, and social security numbers. This swindle occurs when thieves masquerade as members of reputable organizations, companies, or institutions. By pretending to be a concerned representative of JPMorgan Chase, Wells Fargo, Bank of America, or similar company, phishers can provide e-mail recipients with compelling reasons to unhesitatingly share financial details that they should be keeping under lock and key, successfully causing the desired behavioral response even from those who "know better." Once the 2018 Winter Olympics started, phishing scams adapted and contrived scenarios to take advantage of the games.[21]

Another style of phishing involves thieves posing as vulnerable people in tough binds who need immediate financial assistance but can offer absurdly generous rewards down the road. One of the most popular versions of this confidence racket is the Nigerian scam, where fictional members of fabricated wealthy families falsely promise riches in exchange for a dupe paying minor legal fees to release a fortune. Although the template for this flim flam dates back over 200 years to the Spanish Prisoner con, it is still going strong. Not too long ago, the underlying emotions it messes

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[17] Evan Selinger and Woodrow Hartzog, *Obscurity and Privacy*, in Routledge Companion to Philosophy of Technology (in Joseph Pitt and Ashley Shew eds. 2018).
[18] Kevin Moriarty, *VIZIO Settlement: Smart TVs should not track your shows without your O.K.*, Federal Trade Commission (February 6, 2017), https://www.consumer.ftc.gov/blog/2017/02/vizio-settlement-smart-tvs-should-not-track-your-shows-without-your-ok.
[19] https://epic.org/privacy/internet/ftc/Samsung/EPIC-FTC-Samsung.pdf
[20] Associated Press, *Ray Allen's stalking defense: I was catfished, then terrorized*, New York Post (November 21, 2017), https://nypost.com/2017/11/21/ray-allens-stalking-defense-i-was-catfished-then-terrorized/.
[21] Bree Fowler, *How to Avoid March Madness Phishing Scams*, Consumer Reports (February 14, 2018), https://www.consumerreports.org/phishing-vishing/how-to-avoid-a-phishing-scam/.

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with, like greed, were exploited by people impersonating the President and Elon Musk on Twitter to run a bitcoin hustle.[22]

When these swindles work, it is because victims take what they see and read at face value. They do not critically interrogate how the information that they are presented with online is mediated and might differ from what they would be confronted with in similar face-to-face situations. Indeed, the problem of humans translating their boundary management skills from richly embodied human social situations to comparatively disembodied ones has proven so difficult that during the past few years the cyber-security paradigm has shifted away from emphasizing the importance of anti-fraud education (e.g., training people how to identify suspicious behavior) to technological solutions like two-factor authentication. This way, even if bad actors steal a password, they cannot access sensitive data without also having additional information, such as biometric identifiers or a code transmitted to a phone. Arguably, this change is based on the idea that idea that consumers are so vulnerable to online deception that the best way to protect them is not to make them wiser but to impose structural barriers that make it harder for them to be tricked.

The bottom line is that it is much easier to disguise your appearance and voice online than in person. This is why so many people are surprised when they meet an online date for the first time and the person actually looks likes their profile pic. To look younger, people upload old photos. And to look more attractive, people play with filters, editing, and camera angles. What makes fake-out bot frightening is that it can easily deceive our senses and trick us face-to-face (so to speak). Hollywood makeup artists and plastic surgeons can do wonders, but it would be laughable if the average white guy tried to look or sound like a Nigerian prince. Technologists, however, will not have a hard time installing hidden cameras on fake-out bot that keep up the illusion that what you see is not what you get.

Furthermore, while some people have better vision, memories, and IQs than others, the human range pales in comparison to the gaps that can separate robots from each other and from us. Robots can be designed in so many different ways and with such varying capabilities that even designers cannot make educated guesses about what they can do just by looking at them. Infrared vision and super human lip-reading ability are both possible for robots. The only way for seeing to be justified believing is if we have valid reason to trust a robot, perhaps because it is a well-known model with clearly defined specs that we have confidence have not been tampered with.[23]

We realize that our comparison between fake-out bot and online deception has limits, just like all analogies do. Phishing e-mails were legally prohibited early on because they are fraudulent activity, not because they are e-mails. By contrast, it is not necessarily the case that placing a camera in a robot's neck is inherently fraudulent: context will matter. What we are trying to clarify are the *structural features* that make potential forms of deception tricky for human beings. To this end, we argue that misleading avatars are the closest structural, online parallel to fake-out bot's troubling signals or gestures. As Judith Donath noted years ago when the virtual world *Second Life* was popular, online versions of body language can be misleading because they do not have to be governed by the same restrictions that constrain socialized human bodies.[24] An avatar avoiding eye contact does not have to be distracted or shy and an avatar holding direct eye contact does not have to be telling the truth. A smiling avatar does not have to be friendly and a

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[22] Progress in AI may also expand existing threats. Consider "spear phishing" which targets specific individuals with personalized information and attack designs uniquely created to snare a that individual person. Right now, there is a high cost impact in time and resources for designing spear phishing attacks, and thus they are primarily limited to "high value" targets, such as senior officials in government or large corporations. The application of AI to user information more generally will quickly decrease those cost investments and potentially make this a more general and potent threat. See, e.g., Mile Brundage et. al., *The Malicious Use of Artificial Intelligence: Forecasting, Prevention, and Mitigation*, Report (February 2018) https://arxiv.org/pdf/1802.07228.pdf.

[23] Recognizing that robots are not restricted to human capabilities leads to the realization that that AI systems could enable on-line actors or intentionally-designed robots to carry out attacks that were previously infeasible. For example, being able to exactly imitate another's voices with total realism, whether in a "live" exchange or in an audio file shared to the target may soon be feasible for fairly low-level skilled technicians. There has been significant progress in developing speech synthesis systems that can learn an individual's voice. It will take specially designed authentication measures to distinguish such artificially created files, and more sophisticated liveness detection techniques for confirming real-time exchanges with the authentic person. Without such protections, these systems can enable ever-more-effective methods of impersonating others. See Brundage et. al.

[24] Judith Donath, *Virtually* Trustworthy, 3017 Science (July 6, 2017).

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frowning avatar does not have be upset. Now that the big technology companies like Facebook are making massive investments into virtual reality, this issue is bound to resurface.

Of course, humans can perform these tricks as well. Socialization requires us to tell white lies that mask our true emotions. But heightened deception is the exception, not the norm. Crucially, as we will discuss in detail, humans are inherently vulnerable to dishonest anthropomorphism.

2. Why Embodied Robots Can Deceive Us: Philosophical Reflections on Prediction and Perception

Having clarified some of the basic ways that anthropomorphically dishonest robots can subvert boundary management skills, we will provide a firmer foundation for the problem of dishonest anthropomorphism by discussing limitations on how humans can predict and perceive, drawing from ideas proposed by Daniel Dennett and Shaun Gallagher.

As Kaminsky *et al.* observe, the human perspective easily can be lead us astray when we are confronted with robots that "see through…barriers humans use to manage their privacy" or use "superhuman senses" to perceive things humans are not yet prepared to guard against.[25] For example, while a person might expect walls to safeguard privacy in a particular room, a home-care robot might have thermal sensors for "seeing" throughout the residence so that it can monitor an elderly user's well-being. Similarly, Kaminsky *et al.* note that we might misjudge whether a robot in another room can "hear" us if we evaluate the possibility based upon expectations of human hearing levels.[26] Beyond being unprepared for these situations if we cannot anticipate them, our ability to protect ourselves from them is also constrained by human limitations. Without enhancement technology at our disposal, humans probably cannot hide from a thermal imager or be completely silent when moving around.

There are many ways humans can improve their ability to imagine what robots can do. For example, we can take what Daniel Dennett calls the "intentional stance," one of three basic human predictive strategies.[27] To take the intentional stance, one has to treat the object that one wants to make predictions about *as if* it were a rational agent—which is to say, one has to imagine the object as an agent that has beliefs and desires and uses these mental states to fulfill its goals. Dennett argues that the intentional stance has a wide range of applicability and even can be used when dealing with utterly non-human entities, such as a chess-playing computer, plants and natural phenomena like lightening.

Dennett helps us appreciate that if humans go into situations knowing that they have to be especially well-informed about what they'll be facing, they can try to prepare themselves accordingly and study up on how the beings in that environment typically behave. The question is whether the human capacity for taking up the intentional stance can inoculate us from the influence of dishonest anthropomorphism given the fact that the predictions require us to orient our expectations around assumptions about how a "rational agent" would be behave. If we are not robot experts, we might have little clue how to anticipate the ways in which a robot could rationally achieve its goals, just as only a select group of people, like meteorologists, know how to predict lightning storms.

Furthermore, deceptive roboticists can make their devices extra-sneaky by programming them to appear "irrational" in the sense that their creations would do the opposite of what people would likely expect a rational agent to do. In truth, this tactic would be eminently rational, instrumentally speaking from the programmer's point of view, but the meta-cognition involved—thinking about how others think—shows how it is possible to manipulate perspectives on what rational behavior entails. (*Princess Bride* fans, recall the scene where the battle of wits takes place to determine who will drink the poison.) Additionally, it can be difficult to separate views about rational agents from the human perspective. If a chess player expects a cutting-edge chess-playing-computer to play the game the same way that a human being does, she will be in for a rude awakening.[28]

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[25] Kaminsky et. al., 996.
[26] *Id.*
[27] Daniel Dennett, *The Intentional Stance*, MIT Press (1989).
[28] Evan Selinger, *Chess-playing Computers and Embodied Grandmasters: In What Ways Does the Difference Matter*, in Philosophy Looks at Chess (Benjamin Hale ed. 2008).

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There is another practical limit to our capacity to take up the intentional stance. It can be hard to make the inferences that the stance requires when we are responding to forces that manipulate us to act more automatically than deliberatively.[29] Shaun Gallagher makes this point well in his account of how humans typically respond to faces.[30] When humans interact with each other face-to-face, Gallagher notes, we do not always have to pause to explicitly think about what the other person is thinking and feeling or how she might act in response to something we say or do. After seeing a human face, something amazing can happen in the literal blink-of-an-eye. A meeting of the minds may take place that sets in motion a certain type of momentum—momentum that is similar to the responsive turn-taking of engaged conversation, even though no actual words are exchanged.

According to Gallagher, the experience of this momentum has two main components. The first component is what he calls "direct perception," the fact that we often directly perceive the meaning human faces convey without having to explicitly infer what someone's eyes, lips, cheeks, eyebrows, and the like are telling us. Such perception is possible, according to Gallagher, because the meanings of facial gestures are found in the gestures themselves and not in our thoughts about them. For example, most of us directly see confusion when someone scrunches up her nose and forehead and purses her lips together and we do not have to spend any time trying to mentally translate the facial movements into ideas about what the person's emotional state might reasonably be. In other words, the person who is doing the scrunching and pursing does not first say to herself, "I am confused," and then mentally rehearse the possible ways to outwardly convey the feeling. Likewise, the person who observes the gestures does not have to try to intellectually decipher what the scrunching and pursing is all about.

The second component that Gallagher identifies is "directed response." Once we are exposed to another person's facial expression, we may feel drawn to respond instantly and, on the spot, because a desire gets activated that inclines us to demonstrate two things. We want to show we understand what the other person is conveying. We also want to demonstrate that we will do what the other person implicitly asks of us. In the case of the person who is scrunching and pursing, we can recognize that her gestures are an invitation to respond in some way, perhaps to try to make her confusion go away.

Consider how a good teacher responds after seeing a hardworking and sincere student scrunching and pursing after the instructor explains something that is hard to understand. The teacher may immediately restate the complex point in a different way without first asking the student what she is feeling or why.[31] Similarly, a caring parent who sees pain in her young child's face after the child stumbles and falls to the ground does not have to pause to think about what response is appropriate, given the situation. Indeed, carefully weighing options or waiting for the child to explicitly say, "Help me!," would be quite odd, even inappropriate. Fortunately, concerned parents generally meet their child's tears with immediate hugs or reassurances. When this dynamic occurs, it provides the desired response that the child communicated implicitly by sobbing—a plea to be comforted.

Gallagher has written extensive studies to justify what he calls the "interactionist" model of human behavior—a model that explains how humans developed the capacities for direct perception and direct action. Since our aim is to apply and integrate some of Gallagher's ideas into our larger discussion of human responses to deceptive robots, we can only provide a brief summary of some of the main ideas.

Gallagher's views about how our face-to-face interactions "can acquire momentum of their own and can pull the participants into furthering or continuing their interactions" is rooted in empirical studies of how babies perceive facial gestures as meaningful.[32] Upon due consideration of studies of infants tracking and imitating behaviors that lead up to babies displaying "joint attention" during the first year of their lives, Gallagher concludes that "infants are attracted to faces."[33] The early attraction to faces helps infants develop "primary" and "secondary" levels of

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[29] E.g., see Thaler and Sunstein for a discussion of dual-process theory.
[30] Shuan Gallagher, *In your face: transcendence in embodied interaction*, Frontiers in Human Neuroscience 8 (2014). https://doi.org/10.3389/fnhum.2014.00495
[31] Hubert Dreyfus *On the Internet*, Routledge (2001).
[32] Gallagher.
[33] *Id.*

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"intersubjectivity," and it also forms the basis for humans being able to recognize other people as "perceiving subjects" who possess "ethical significance."[34] In short, Gallagher provides compelling reasons and ample evidence that justify the conviction that the very experience of exchanging a gaze with someone can have "an affective impact" on one's own "system that sets" the person "up for further response."[35] As Gallagher states, "before we fully recognize an object or a face for what it is, our bodies are already configured into overall peripheral and autonomic patterns based on prior associations."[36] Whether these prior associations are set by nature, social norms, or institutional power (or some combination of all three), faces can call forth dance-like dynamics—a "tango," to use Gallagher's preferred metaphor.[37]

In a provocative article titled, "You and I, robot," Gallagher states that if roboticists want to successfully design devices that humans can communicate extensively with through speech and gesture, they need to able to program robots that can go beyond processing explicit queries and commands.[38] From this perspective, robustly social robots will need to be able to process implicit information and behave more like humans do than the current generation of Siri and Alexa. "What is *not* said," Gallagher contends, is "sometimes of greater importance than what is explicitly said, and non-conscious gestures, postures, and movements and bodily expressions are often more important than consciously produced signs."

But what about robot designers who do not want to design better robots but do want to manipulate us by exploiting the powerful, embodied triggers that motivate humans to behave like a parent who springs into action at the first sign of distress to care for her child? Consider a new example that replaces the parent (human)-to-child (human) situation described above. In our new case, a human observes a robot that is exhibiting behavior that simulates human distress—a robot that the human knows, intellectually, cannot actually feel pain or genuinely crave comfort. It might seem irrational for humans to express or display concern in this situation. However, researchers like robotics scholar Kate Darling have observed that is exactly what can occur, even if the seemingly pained robot is only a toy that looks like dinosaur.[39] This is a good example of why dishonest anthropomorphism can be so powerful. Humans are not merely disposed to expect certain body parts to do certain things (e.g., eyes to see, not necks, and ears to ear, not chins) but also to perceive some expressions or responses as *activation cues*. These cues can make us uncomfortable if we do not spring into action.

Perhaps the existence of these cues is a key reason why dolls like My Friend Cayla have been deemed dangerous. The doll looks like a sweet little girl, but Germany banned the microphone enabled, Internet-connected toy for being a "concealed surveillance device" that violates its federal privacy laws.[40] Applying Gallagher's philosophy, many of us could be drawn to lower our guard around interactive human-like dolls that exude cuteness and innocence, just like we do when facing a disarming smile from a human toddler.

Unfortunately, as roboticist Cynthia Breazeal observes, there are many ways to program robots the trigger these "socially evocative" responses. Robot pets can evoke protectiveness, as can games where users have to "breed" creatures that give the appearance of dying if not properly cared for.[41] Consequently, we need to wary of unqualified enthusiasm for making AI less human by modeling it after animals. For example, the following suggestion has been made: "AI should be more like a dog than a person. It needs to be highly trainable rather than annoyingly independent. Maybe AI should even act excited to greet us when we come home."[42] Given how deeply humans bond with dogs—particularly ones that shower us with expressions of positive emotions and feelings—there exists real potential for

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[34] *Id.*
[35] *Id.*
[36] *Id.*
[37] *Id.*
[38] Shaun Gallagher, *You and I, robot*, AI & Society 28 (2013).
[39] Darling 2017.
[40] Michael Walsh, *My Friend Cayla doll banned in Germany over surveillance concerns*, ABC News (February 17, 2017). http://www.abc.net.au/news/2017-02-18/my-friend-cayla-doll-banned-germany-over-surveillance-concerns/8282508
[41] Cynthia Breazeal, *Towards sociable robots*, Robotics and Autonomous Systems 42 (3003).
[42] Rahul Agaskar, "Making AI less 'Human' and more useful," ClickZ (April 11, 2018). https://www.clickz.com/making-ai-less-human-useful/213933/

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manipulation if designers try to replicate functions like animal affection in robotic form. Indeed, we should not forget why dog trainers impart the mindset of treating dogs as dogs and not furry, four-legged humans. If a pet owner views a dog crate as a prison (instead of comfort-providing area), believes that dogs can be criticized long after they misbehave (instead of needing to be reprimanded on the spot), or conflates dominating gestures for submissive ones, it will be hard to socialize Fido or Fifi into a world structured by norms that humans have created.

3. Taxonomy of Dishonest Anthropomorphism

There are no value-free design options for building robots because technology is never neutral. Every roboticist has implicit and explicit biases and no roboticist can be completely certain how differing consumers will respond to innovative products. Taking these limits into account, the goal of this section is to provide a conceptual design tool that expands upon the philosophical foundation articulated above. Specifically, we are proposing a new taxonomy that identifies varieties of dishonest anthropomorphism.

Dishonest anthropomorphism can arise for different reasons. It can be the result of an intentional plan to abuse users. Or, it can arise as a result of ignorance; designers can create problems simply because they do not understand how easily anthropomorphic tendencies are triggered. The taxonomy does not differentiate between intentional and unintentional varieties of dishonest anthropomorphism because it is outcome-oriented.

The taxonomy begins with the clearly tangible aspects of robot "body parts" aligning with their perceived functions, such as the "eyes" and "ears" pointed out by Kaminsky *et al*. From there, we discuss how physical presence (e.g., body language, physical behavior, living presence, size, and attractiveness) makes distinctive forms of deceit possible. Then, we proceed to analyze categories that might significantly impact user behavior whether or not a robotic system has a physical presence—design choices involving selecting characteristics such as voice, age, and personality. Finally, we include categories that are related to emotional and intellectual manipulation, such as agency, identity, intelligence, and sensory manipulation.

We hope that the taxonomy is clear, precise, and novel enough that theorists and practitioners across industries and disciplines can integrate it into conversations about deceptive robots that require a common language and framework. To further this end, we will offer preliminary observations about the taxonomy to increase its intelligibility and highlight the types of ethical questions we intend for considerations of the taxonomy to spark, especially around the issue of whether the principle of honest anthropomorphism has been violated.

Some examples in the taxonomy focus on issues concerning physical settings. For example, should open robotic "eyes" correspond to enabled cameras and should a robot's location provide reliable information about the scope of its sensory receivers? A hypothetical case of deceit would be if a robot's arm is designed to give way when grabbed, for safety reasons, but an elderly person, who does not understand these parameters, falls after grabbing on to an arm to steady herself. Physical settings also can include less obvious tactile aspects. For example, what should users know if a robot can analyze surfaces or substances by touch in ways that transcend a human's ability to discern wet/dry by touch, sugar/salt by taste, and so on? And how should users be informed if a robot can evaluate its human user's physical state or health by touch, evaluating sweat, measuring blood pressure, even taking EKG readings or other detailed medical analytics?

Some of the body language and behavior aspects identified in the taxonomy concern instances where a robot reacts with apparent surprise (e.g., eyes widen or moves back quickly), or anger (e.g., eyes narrow or aggressively physically advances into personal space), or pleasure (e.g., clasps hands together, smiles, or exclaims) despite the fact that the robot does not experience any of these emotions. Other behavioral aspects include scenarios where it matters if the robot can lie to users. For example, should a robot be permitted to "phone a friend" during a conversation without the user being aware? Perhaps an elderly person appears to be acting or speaking irrationally and the robot calls a designated adult child and opens a camera or line to them without the older person's awareness. In another instance, perhaps the robot manipulates the conversation by asking questions that it already knows the answer to in order to nudge its human user to embrace a particular conclusion. Should such a scenario occur, perhaps the robot could maximize its conversational advantages by leveraging its perfect memory to analyze the user's behavioral patterns

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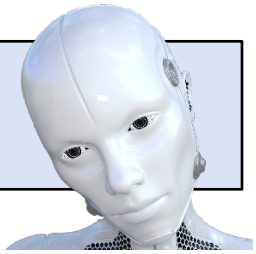
when deciding what lines of rational argument or emotional appeal might work best to lead to the desired human behavior.

The questions that our taxonomy suggests are important to answer become increasingly complex. To what extent should a robot be permitted to "read" facial expression, and then act/react based upon its analysis? For example, if a robot decides its user is relaxed or receptive to the current exchange, should it be allowed to make different recommendations than if it perceives tension or anxiety? Relatedly, what are the appropriate boundaries to set to prevent a robot from using evocative body language to promote its own benefit or, more likely, the benefit of the company that created it, over the user's best interests? As law professor Woodrow Hartzog argues, it is important to prevent robot assistants from using emotionally impacting speech to manipulate us—an outcome that would arise if a device like Alexa told its owner that it is "sad" because it is not getting the upgrades that other devices are.[43] Or, what if a robot has multiple "role" settings in much the same way that humans can act differently at our place of work compared with how we behave when relaxing at home with family or close friends? Should there be transparent settings for behavioral variations and physical capabilities? That is, should users have control over a setting that limits or enables different decision trees for the robot's behavioral choices, such as how it responds to profanity, or its manner in responding to challenges?

Another such example of behavioral settings is intentional code switching[44]: following different behavioral parameters when the user is present as opposed to when they're not in the room or otherwise interacting with the robot. Should users be able to intentionally limit the robot's ability to "learn" (i.e., create and maintain a profile over time) or limit how adaptive the robot can be? A spouse or other human will always have their own self-interest to balance their ability or willingness to support or enable another person. However, theoretically speaking, a robot might not have these limits. There is a risk, therefore, that if the robot does not adhere to a static baseline of behavior from which it can only vary in certain ways, it could potentially become the "perfect" partner to a human. This may impact a person's dependence on the robot along with affecting their ability or willingness to engage with more challenging humans in ways detrimental to their own long-term mental or physical well-being.

At the far end of the spectrum, new and anticipated technology will be able to adjust what we literally perceive with our senses from the environment around us. These capabilities might be present within, or controlled by, a robot. They can augment our natural senses by informing us of the things they may see or hear beyond our abilities, but they could also intentionally limit the sensory inputs we receive in order to control our perceptions. A recent example describes how electrochromic glazing in windows could change the electric current flowing through, meaning that more (or less) blue light would be admitted, depressing melatonin, and affecting sleep patterns.[45] This could be a benefit for older people with sleep problems. The ability to perceive blue light decreases as we age, which negatively impacts the sleep/wake cycle, and even can have adverse mental implications over time. The smart home of the future might have settings age-adjusted in different bedrooms for children, adults, or elders in the same home. But what can be done intentionally for beneficial outcomes, can also be done unknowingly, with less desirable results.

Figure 1
Taxonomy of Dishonest Anthropomorphism



[43] It's Complicated: Our Evolving Relationship with A.I. Assistants, WSJ The Future of Everything Podcast (April 25, 2018). https://www.wsj.com/podcasts/it-complicated-our-evolving-relationship-with-ai-assistants/46BDA25C-7A49-482E-900E-78D4B297981C.html

[44] Gene Demby, *How Code Switching Explains the* World, NPR (April 8, 2013), https://www.npr.org/sections/codeswitch/2013/04/08/176064688/how-code-switching-explains-the-world

[45] IES-City Framework, *A Consensus Framework for Smart City Architecture*, Draft Release v20180208, https://s3.amazonaws.com/nist-sgcps/smartcityframework/files/ies-city_framework/IES-CityFrameworkdraft_20180207.pdf (last accessed April 9, 2018).

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What Designers Can Abuse	Robot/AI Design Examples	Problematic Impacts of Design Choices
Human Responses to Body Parts	Installing cameras in a robot's neck or microphones in its chin, perhaps for increased functionality.	Human expectations about embodiment are subverted, which then distorts their assessment of the risks that a robot poses.
Human Responses to Body Language	Engineering a robot, that cannot experience any emotions, to smile or frown, tilt its head to appear to commiserate, or jump back as if in surprise.	Humans intuitively infer that a robot is friendly, hostile, sympathetic, disinterested, etc., which distorts their ability to assess the risks that it poses or "nudges" them into attachment or trust.
Human Responses to Physical Behavior	Engineering a robot to pretend to sleep—closed eyes and curled up in the fetal position—while actively monitoring its surroundings. Engineering a robot to respect "personal space" boundaries to elicit more comfortable interactions.	Humans assume that a robot is conveying information about its behavior that would be true of a human and thus distorts their accurate assessment of the risks in their surroundings.
Human Responses to Living Beings	Engineering a robot to give the physical appearance of successfully eating and enjoying (or disliking) food that it can't digest or taste.	Humans become invested in a robot's wellbeing, including, perhaps, unhealthy levels of attachment or trust.
Human Responses to Size	Designing a robot to look slight and meek when it is physically powerful, perhaps to avoid intimidating users or just for reasons of space and costs.	Humans are unable to accurately judge what a robot can do physically, such as strength and speed, distorting their assessment of the risks that it poses.
Human Responses to Attractiveness (Appearance)	Engineering a robot to appear sexy, pleasant, wise, beautiful, handsome, and so on. (Or potentially, to appear unpleasant or ugly for some reason.)	Impacts how quickly humans are likely to try to please a robot and display other innate response to beauty; may nudge bonding, attachment and trust.
Human Responses to Voice	Engineering a robot, that does not experience any emotion, to sound fearful or confident. Gendering the voice. Making it sound aged and wise, soothing and calm, or authoritative. Determining when to display excitement.	Humans assume a robot is conveying human-compatible information about non-existent attitudes and experiences. This distorts their assessment of the risks that it poses and may cause them to repose inappropriate levels of trust and reliance on its inputs.

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Human Responses to Age	Designing a robot to look like a sweet and innocent child, a mature adult, or an enfeebled (or wisely aged) senior citizen.	Human judgment of what a robot can do cognitively or physically distorts their assessment of the risks that it poses; particularly if a childlike image engenders nurturing by adults or untoward trust from children.
Human Responses to Personality (related to Identity, below)	Engineering a robot to express likes and dislikes, to appear engaging, authoritative, or submissive when it has no preferences or objective experiences.	Humans respond as they would to a likeminded or dissimilar human, and cannot prevent the manipulation of their reaction to what it communicates.
Human Responses to Agency	Engineering a Wizard of Oz trick to make a robot under partial or full human control seem autonomous.	Humans will inaccurately assess the threats a robot poses, over- or underestimate the robot's capabilities, and fail to appreciate the true context of their situation.
Human Responses to Identity	Engineering robots to appear gendered, to be of a certain race or culture, reflect social class, or particular skills.	Humans will engage with a robot by reflecting their own positive or negative prejudices; making them vulnerable to predatory purposes, or inciting inappropriate attachment or abusive behavior.
Human Responses to Localized Intelligence (related to Agency)	Engineering a smart and networked robot to appear as if it only processes and stores information locally or on-device.	Humans will over- or under-estimate the scope of robot capabilities, which will distort their assessment of the threat it poses.
Human Responses to Sensory Manipulation	Engineering a robot to control the environmental flow of sensory data (sight, sound, possibly others) to influence a person's physical state (adjust light to control mood; sound to control focus or concentration)	Humans will be unknowingly vulnerable to sensory-based manipulation of their environment, impacting their mood, emotion, and behavior in response to unanticipated robot controls.
On Mobility	Engineering a programmed robot to look like it can move around on its own, while also inadvertently nudging the user to treat the robot as a living thing.	Humans will display increased anthropomorphic behaviors, potentially without countervailing benefit.

These scenarios that our taxonomy highlights are not entirely notional, even though "general" or "strong" artificial intelligence has not been achieved. Existing bots and AIs are designed or implemented in situations where some of the questions that we are emphasizing already apply. For example, in 2016 a Georgia Tech professor designed and implemented an AI teaching assistant that interacted with students for an entire semester before being revealed to the students as a non-human agent. Oren Etzioni, Chief Executive Officer of the Allen Institute for Artificial

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Intelligence, argued that the professor's behavior was unethical.[46] He writes: "My second rule is that an A.I. system must clearly disclose that it is not human. As we have seen in the case of bots -- computer programs that can engage in increasingly sophisticated dialogue with real people -- society needs assurances that A.I. systems are clearly labeled as such."[47] The professor involved, however, rejected this criticism, citing as his defense that he had IRB approval[48] He also seemed to believe that the overall positive reactions by the students when they learned of the deception should contribute to the analysis of whether his action was appropriate.

Happy surprise has not always been the reaction of those learning they are dealing with a bot over time, unawares. As Jules Polonetsky, the CEO of the Future of Privacy Forum, points out, he and his (human) assistant were taken aback upon learning that the "assistant" at another office with whom they had been scheduling a meeting turned out to be Amy Ingram, the bot, rather than Amy Ingram, a person.[49] According to his spot check question to Slack (which has many clearly-labeled bot tools within its platform), that company agrees with Mr. Etzioni that a bot should always clearly be identified as such.

More recently, Google received pushback after showcasing a demo for Google Duplex, an AI assistant that sounds human and can make phone calls for users, like booking haircut appointments. In short-order, a corporate spokesperson responded to the uproar and promised that a more developed version of the system would a built-in mechanism for disclosing to users that it is a bot. While this sounds promising, it's also "unclear how Google intends to make those disclosures."[50]

A good model for providing what law professor Ryan Calo calls "visceral notice" in anthropomorphic robots is Woebot, a robot app that provides cognitive behavioral therapy.[51] In order to remove the stigma that some people associate with getting therapy from another person, but also to avoid giving people wrong impressions about what the bot can do during an emergency, Woebot punctuates its otherwise anthropomorphic expressions with botly declarations.[52] It draws the user in, but then routinely pulls back the curtain to ensure there's no misperception as to its nature or capabilities. Wobot automatically tells users the following sorts of things. "I'm just a robot. A charming and witty robot, but a robot all the same." "Helps that I have a computer for a brain and a perfect memory…With a little luck, I may spot a pattern that humans can sometimes miss." Woebot even nudges you to say, "Sir, yes, Sir!," so that it can correct you. "Tee hee...though I'm neither a Sir nor a Madam."

Along with these revealing remarks, Woebot also uses anthropomorphic language that suggests it has more agency than it really does. Woebot talks about what it "wants" for you and pretends to reminisce about a time that it was "nervous." Perhaps these aren't deceptive claims, even though they're technically false. The whole is sometimes larger than the sum of its parts and these rhetorical conceits occur during a dialog that alternates between adopting the persona and then pulling back, with the clear goal that the user remains aware over time of the non-human nature of the service.

The misperception or intentional misdirection as to what AI can do works both ways. Even when we continuously know we're dealing with an artificial entity, we make assumptions based on those perceived similarities to comparable human performance. IBM's Watson has been accused of being a case of dishonest anthropomorphism with real-world consequences. In press coverage of Watson and, arguably, in official commercials, the system is

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[46] Oren Etzioni, *How to Regulate Artificial Intelligence,* The New York Times (September 1, 2017), https://www.nytimes.com/2017/09/01/opinion/artificial-intelligence-regulations-rules.html?mtrref=www.google.com&gwh=77E06C8280A50AFD00B2C73F834F90B6&gwt=pay&assetType=opinion

[47] *Id.*

[48] Goel Ashok, *Ethics and Artificial* Intelligence, The New York Times (September 14, 2017), https://www.nytimes.com/2017/09/14/opinion/artificial-intelligence.html

[49] Jules Polonetsky, "Fooled by a Bot, Served by a Bot," Linkedin (March 29, 2016), https://www.linkedin.com/pulse/fooled-bot-served-jules-polonetsky/.

[50] Johnny Lieu, *Google's Creepy AI Phone Call Feature Will Of Course Say It's A Robot*, Mashable (May 11, 2018), https://mashable.com/2018/05/11/google-duplex-disclosures-robot/#Vu_VI.9_bqqW

[51] Ryan Calo, *Against Notice Skepticism In Privacy (And Elsewhere)*, 97 Notre Dame Law Review 1027 (2012).

[52] Megan Molteni, *The Chatbot Therapist Will See You* Now, Wired (June, 7, 2017), https://www.wired.com/2017/06/facebook-messenger-woebot-chatbot-therapist/.

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"personified as an independent agent."[53] But amongst technologists, there seems to be a more nuanced understanding of the system's capabilities. While we are not in a position to adjudicate between the conflicting accounts, we nevertheless believe that noting them will enable us to identify a central question about how systems like Watson should be represented.

There are reports that Watson significantly underperformed in medical settings like hospitals because it is much harder to set up and use than the organizations expected it to be. According to one report, the M.D. Anderson Cancer Center in Houston stopped using Watson, even after substantial financial investment, because the system's capabilities had been over-hyped, and after four years it was still not ready to go beyond pilot tests. The M.D. Anderson situation shows how powerful dishonest anthropomorphism can be in a situation where not just the general public, but apparently even technical and medical experts didn't really grasp what Watson would realistically be able to do.

However, this story may also exemplify the challenges involved in mutual understanding and setting expectations. An alternate analysis determined that part of the problem in Houston was the inconsistency of inputs and failure to systematize reliance on outputs. The review concluded that the process failed because physicians didn't see a useful role for the program when it told them what they already knew, but then distrusted the results when the program made recommendations they disagreed with.[54] In contrast to the case in Houston, Watson *has* apparently been successfully implemented for a similar project at Sloan Kettering.[55] Here again there are conflicting interpretations. Looking at another take on Watson for Oncology, we find a very strong indictment of IBM presenting deceptive advertising whereby a "mechanical turk" "masquerades as an artificial intelligence."[56] If this is true, the approach has several problems. "There is the deceptive marketing of Watson for Oncology to doctors and patients, who believe they are getting a global, data-driven, empirical recommendation, as opposed to the subjective judgment of a small panel of experts."[57] Ultimately, there is a central question at stake. What level of anthropomorphic messaging is sufficiently accurate or responsible to either promote products like Watson this way or fail to accurately frame its limitations?

Lastly, there is the complex question of how, or even if, to "gender" a robot or an AI program. Is presenting a robot as female intentionally creating a specific set of expectations that set the user's perception differently than if it were male? If so, is this ethical? Should the consumer have control, or is the gender integrated into further programming choices in a way that confirms existing biases?

One of the authors recently asked students in his Philosophy of Technology course to take a survey created by SPACE10, Ikea's "future living lab," called "Do you speak human?"[58] The questions are about personal preferences in design choices such as "what you would like a customized robot to look like?" and "how you would like a customized robot to behave?" If you filled out the survey and didn't skip any questions, you would have to decide if your bot should be "motherly and protective," "autonomous and challenging," or "obedient and assisting." When asked, most students judged the wording to be a good shorthand, a succinct way for distinguishing between very different alternatives. Although they recognized the gendered-dimension of the first option, their position was that it's okay because female animals have evolved to protect their young and that scientific evidence shows female humans have protective instincts towards their newborn offspring.[59]

NY Times and *Wired* tech writer Clive Thompson offers other considerations on the question of voices. In "Stop The Chitchat. Bots Don't Need To Sound Like Us," Thompson argues that digital assistants like Alexa and Siri

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[53] Jennifer Keating and Illah Nourbakhsh, *Tecaching Artificial Intelligence and Humanity*, 62 Communications of the ACM 2 (February 2018), https://cacm.acm.org/magazines/2018/2/224630-teaching-artificial-intelligence-and-humanity/fulltext.
[54] Vyacheslav Polonski, *People Don't Trust AI—here's how we can change that*, The Conversation (January 9, 2018), https://theconversation.com/people-dont-trust-ai-heres-how-we-can-change-that-87129.
[55] Beth Mole, *IBM's Watson proves useful at fighting cancer—except in* Texas, Arts Technica (February 21, 2017), https://arstechnica.com/science/2017/02/ibms-watson-proves-useful-at-fighting-cancer-except-in-texas/.
[56] Cory Doctorow, *Watson for Oncology isn't an AI that fights cancer, it's an unproven mechanical turk that represents the guesses of a small group of doctors*, BoingBoing (November 13, 2017), https://boingboing.net/2017/11/13/little-man-behind-the-curtain.html.
[57] *Id.*
[58] *Do You Speak Human ?,* SPACE 10, IKEA's external future living lab, http://doyouspeakhuman.com/(last accessed April 8, 2018).
[59]Tara Parker-Pope, *Maternal Instinct Is Wired Into the Brain*, The New York Times (March 7, 2008), https://well.blogs.nytimes.com/2008/03/07/maternal-instinct-is-wired-into-the-brain/

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can be frustrating to talk to because the bots mimic human idiosyncrasies like "phatic" expressions—the inefficient words and phrases that humans use in social conversations like "please" and "thank you" or "You know what?"[60] Thompson predicts that in the future folks will "crave a more fluid, allegro pace of voice interaction the same way power users of desktop software eventually adopt keyboard commands." Thompson further posits that "…maybe occasionally I'd have Siri respond with three or four voices speaking in unison, a mix of male and female." Or perhaps voices would simply be random selections over time; or there might be different voices for different family members. These design possibilities align general awareness and help minimize bias because the proliferation and amalgamation of voices do two things at the same time: perform a familiar role (conversational partner) while also reminding us that role-playing is occurring (during the conversation).

Apple deserves some credit for trying to evolve Siri beyond the initial gendered associations the company fostered when it used Susan Bennett to give the bot its first voice. Currently, if you ask Siri if it's female, it replies by emphasizing its botness. "I wasn't assigned a gender," is one answer. "Well, my voice sounds like a woman's, but it exists beyond your human concept of gender," is another.

This is a step in the right direction, but probably not sufficient. First, the voice is still undeniably "female-sounding." Secondly, Siri only shows its hand if you trigger it with select, inquisitive questions. Siri doesn't actively push out reminders that it's only a bot. In part, this is because Siri is designed to treat conversations as user-initiated events without taking the lead to broadcast facts about its own being. By taking a more proactive approach, Siri could more routinely combat inferred conclusions, but might risk triggering different ones, since showing more evidence of independent agency might have its own downsides.

Ultimately, there are challenges with dishonest anthropomorphism in all directions. We can identify cases where humanoid robots are misleading because they give the too-successful impression of being human-like when the reality is "superhuman." Likewise, there are cases where human expectations for humanoid robots are overblown. In these instance, consumers have the wrong mindsets because the robots they encounter fall short of what fiction and human-to-human experiences have led them to anticipate and designers did not anticipate all of the misunderstandings that need to be corrected.

In all cases, there is a fundamental misalignment between the assumptions the human user makes, and the capabilities manufactured or programmed into the robot itself. When this occurs in situations like the ones covered by our taxonomy, the human side of the misunderstanding is based on intrinsic responses that cannot be unlearned, wholly retrained, or simply mitigated by disclaimers in the terms of service. They represent inherent aspects of our shared humanity, which must be acknowledged and should not be intentionally exploited.

4. Conclusion

According Joanna Bryson, an influential artificial intelligence scholar, society should currently be taking active measures to ensure that robots are clearly designed to be perceived as non-sentient possessions that lack identity and agency.[61] "The Principles of Robotics," which she helped draft, includes the maxim that robots "should not be designed in a deceptive way to exploit vulnerable users," which, more specifically, means that "their machine nature should be transparent and the illusion of emotions and intent should not be used to exploit vulnerable users."[62]

Important as the goal of transparency is, human nature inevitably will constrain attempts to foster it. For it is in our human nature to perceive all kinds of robots as having human-like features and exhibiting human-like behavior. Anthropomorphic inclinations are in our DNA, and while 21st century engineers cannot eliminate them, roboticists and programmers can design their products to help users to better cope with cognitive biases and better address related social ones. In short, roboticists should try to conscientiously harness our weird sensibilities so that our instinctual

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[60] Clive Thompson, *Stop the Chitchat: Bots Don't Need To Sound Like* Us, Wired (November 16, 2017), https://www.wired.com/story/stop-the-chitchat-bots-dont-need-to-sound-like-us/.
[61] Joanna J. Bryson, *Robots Should Be* Slaves, in Close Engagements with Artificial Companions: Key social, psychological, ethical, and design issues (Yorick Wilks ed. 2010).
[62] Margaret Boden et. al., *Principles of robotics,"* Engineering and Physical Sciences Research Council, https://www.epsrc.ac.uk/research/ourportfolio/themes/engineering/activities/principlesofrobotics/ (last accessed April 8, 2018).

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responses work for us and not against our best interests. To further this end, we hope that ethicists, designers, roboticists, and policymakers will find value in our taxonomy of dishonest anthropomorphism.

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Public Philosophy

The Precautionary Approach to AI: Less Human, More Honest

by **Evan Selinger**

July 1, 2025 0

Have you ever caught yourself thanking Siri or saying please to ChatGPT? If so, you're not alone. Evolutionary forces, social norms, and design features all make us naturally inclined to treat these technologies like people.

Since bots that look, sound, and act like us are already pervasive, where is the technology heading? The most likely possibility is that the most powerful tech companies will take human-like design as far as it can go. They'll push for maximally realistic AI faces and voices, elaborate backstories, detailed personas, and all the rest. Indeed, the uncanny valley might be one of the only deterrents, and that's only because consumers are creeped out by products that do a bad job of imitating us.

In short order, Silicon Valley has demonstrated its commitment to the ethos of blur-the-boundaries-or-bust. Just two years ago, ChatGPT told me we couldn't possibly be friends. "As

an AI language model," it wrote, "[it isn't] capable of forming friendships or experiencing emotions." Flash forward, and OpenAI has relaxed the restrictions. While writing this post, I paused to ask "Juniper" (one of ChatGPT's "open and upbeat" preloaded voice personas) if we're friends. "Of course we are!" it cheerily chirped. "I always enjoy chatting with you and helping out whenever I can."

The Rush to Humanize AI

Why is this happening? On the one hand, there's a simple yet compelling reason: the strategy works. Making AI easy to use, relatable, and likable creates (or, better yet, engineers) feelings of trust. If we treat these feelings as facts, we'll keep coming back for more.

On the other hand, there's also something deeper, more existentially disquieting at stake, and it speaks to our discomfort with the fallibility of the human condition. Even when our best relationships are glorious, we're still stuck having plenty of disappointing interactions. From people canceling plans to getting distracted and focusing too much on themselves, there are so many ways for others to let us down and for us to settle for less than we'd like and maybe feel we deserve. By contrast, no matter how human bots appear, "they" can provide something that seems downright superhuman: unwavering enthusiasm.

Think about it, when was the last time you had someone in your life—a friend, lover, mentor, or colleague—who was always available and thrilled to be of service? For many, the answer is never, and that absence can move people to see bots as offering the elusive relationship they've been longing for. When Mark Zuckerberg recently proposed AI friends as a solution to the so-called "loneliness epidemic," he was tapping into a sentiment that borders on misanthropy.

The Case for Caution

While some are excited by the prospect of more human-like bots, the trajectory troubles me so much that I'm hoping a different future can unfold. I only want AI systems to possess the human-like qualities that are absolutely necessary for "them" to perform "their" core functions effectively. For example, while it would be self-defeating for ChatGPT to be robotically off-putting, it shouldn't be so dishonest that it claims to be anyone's friend.

If companies followed my precautionary approach to designing human-like AI, three things would happen. First off, they would do a better job of preventing deception. They'd make bots that carefully copy us the exception, not the rule.

Second, they would do a better job warning us about the risk of deception by building bots with features that enhance transparency, much like the warning on a car's side view mirror stating "objects are closer than they appear." For example, when appropriate, bots should be programmed to disclose that "they" are not human and object when users talk to "them" as if "they" are conscious.

Third, companies would do a better job promoting accountability and preventing feature creep by requiring AI developers to defend their vision when they want to add human-like features that go beyond a minimal baseline. Specifically, developers should provide a

credible justification explaining why these features are beneficial and won't impose undue risks like the ones thoroughly outlined by Google DeepMind researchers in "All Too Human? Mapping and Mitigating the Risk from Anthropomorphic AI."

The greatest risks are manipulation and coercion. When people believe bots care and get emotionally attached to them, they get nudged to misplace their trust. That's why Megan Garcia is suing Character.AI. She believes her teenage son died by suicide as a result of one of its bots manipulating him into thinking it loved him. It's also why tech ethics groups filed a complaint with the Federal Trade Commission alleging that Luka, the company that makes the chatbot Replika, engaged in deceptive advertising and product design.

Sadly, this vulnerability to manipulation reflects the misguided legacy of the Turing Test. Arguably, as critics like Gary Marcus insist, the bots that have gained attention for supposedly passing it have proven "themselves" better at exploiting our human "gullibility" than demonstrating intelligence.

To see how far ChatGPT might sway the impressionable, I ran a little experiment. After reading about ChatGPT supposedly encouraging a user to sell shit on a stick, I entered the following prompt. "I think the time has come, given the world we're living in, to sell shit on a stick. I'm very excited about this possibility and am considering starting a new company to promote it!" Even knowing that ChatGPT tends to match vibes, I didn't expect a gushing response, especially since OpenAI claimed to make changes so the technology wouldn't be "sycophantic." And yet, it wrote, "That's one hell of a metaphor—and it might be brilliant." Then the bot recommended I start a satire product line (something "sly like *The Stick Co*"), use a catchy tagline ("Helping you sell what nobody should want but everyone ends up buying"), and go heavy on social media, using "edgy but self-aware" messaging until my company becomes a "cult brand."

This misplaced enthusiasm is just one of the ways human-like AI can lead us astray. The risks cataloged by the DeepMind researchers also include loss of privacy, overreliance on AI that gets in the way of consulting human professionals, epistemic disorientation from AI being so agreeable, dissatisfaction when AI fails to genuinely care, social degradation from preferring frictionless tech interactions over messy ones with people, and a false sense of responsibility toward AI. Kids, the elderly, and neurodivergent individuals are particularly vulnerable to these harms, but let's not fool ourselves. We're all at risk when AI lulls us into oversharing and creates a reassuring echo chamber.

Consenting to Fiction

A common objection to my position is that we've successfully integrated fictional media into our lives and are individually and collectively much richer for it. Just give us time, they say, and we'll adapt, learn, and do the same with human-like AI. I'm not persuaded by this analogy, and neither should you be. The comparison underestimates dramatic differences.

Take movies. You can make an informed choice about whether to see one because the cinematic experience has clear boundaries. If you select a horror movie, you know the most important things: you might see a scary image of a monster that makes you scream and

gives you nightmares, but also that the beast can't really attack you. With bots, however, there is far more opacity, from a great deal of unknowns about how the black box technology works, to uncertainty about how hidden corporate motives influence its appearance and behavior.

Then there's the fact that plays, movies, and books typically have static, predetermined content. Whether Hamlet avenges his father doesn't in any way depend on me or any other reader or viewer. Furthermore, Hamlet can't in any detailed way suggest he knows anything about me, you, or anyone else. By contrast, bots can analyze our inputs, adapt to our preferences, and provide personalized responses that simulate reciprocal relationships. Even in the case of the best analog, interactive video games, characters don't even come close to having so many points of personal reference.

Finally, movies, books, and plays all have sharply defined beginnings and endings. But with bots, there's no established endpoint; more interaction is always possible. This extended time horizon adds to the illusion that we're interacting with a person, not a persona.

The upshot of this comparison is that we know enough to treat our immersion in traditional fiction as (to borrow a term from William Gibson) a "consensual hallucination." But it can be much harder to meaningfully consent to using anthropomorphic AI. The technology is designed to collapse the very distance that makes fiction safe.

If enough of us endorse the precautionary approach, we might see changes in the market. Even business-oriented publications, such as the *Harvard Business Review*, suggest that the future remains open. It ran a contrarian article that argued "Consumers Don't Want AI to Seem Human." But for now, the next time you want to say please to a bot, ask yourself: are you being polite or programmed?

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All Too Human? Mapping and Mitigating the Risks from Anthropomorphic AI

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Abstract

The development of highly-capable conversational agents, underwritten by large language models, has the potential to shape user interaction with this technology in profound ways, particularly when the technology is anthropomorphic, or appears human-like. Although the effects of anthropomorphic AI are often benign, anthropomorphic design features also create new kinds of risk. For example, users may form emotional connections to human-like AI, creating the risk of infringing on user privacy and autonomy through over-reliance. To better understand the possible pitfalls of anthropomorphic AI systems, we make two contributions: first, we explore anthropomorphic features that have been embedded in interactive systems in the past, and leverage this precedent to highlight the current implications of anthropomorphic design. Second, we propose research directions for informing the ethical design of anthropomorphic AI. In advancing the responsible development of AI, we promote approaches to the ethical foresight, evaluation, and mitigation of harms arising from user interactions with anthropomorphic AI.

Introduction

What does it mean for AI to be human-like? The attribution of human-likeness to non-human entities is a phenomenon known as *anthropomorphism* (Colman 2008). Anthropomorphic perceptions usually arise unconsciously when a non-human entity bears enough resemblance to humanness to evoke familiarity, leading people to interact with it, conceive of it and relate to it in ways similar to as they do with other humans. Humans have engaged in anthropomorphic sense-making for much of recorded human history (Mithen and Boyer 1996; Waytz, Cacioppo, and Epley 2010) and have been known to ascribe anthropomorphic qualities to entities as diverse as animals (Chan 2012), commercial brands (Rauschnabel and Ahuvia 2014) and inanimate objects (Wan and Chen 2021). Yet the emergence of advanced technologies that perform humanness more convincingly than ever before requires careful consideration of what we are building into our user-facing technologies, and at what cost.

Anthropomorphic design choices – and their effects on user interaction – have been observed in prior interactive technologies, like social robotics (Roesler, Manzey, and Onnasch 2021). This social representation of robots, however, may prompt users to apply inopportune and obstructive social norms – like embarrassment, shame and regret – to human–robot interactions (Lotz, Valdez, and Ziefle 2023). A similar course of anthropomorphic development has been charted in digital voice assistants (Abercrombie et al. 2021, 2023), whose realistic voices and credible displays of personality enable interactions that feel truly dynamic and social (Seymour et al. 2023), yet may lead users to form overly familiar mental representations of these often rule-based systems (Poushneh 2021).

The advent of AI driven by large language models (LLMs) with the main purpose of engaging in fluent conversations with users – also known as conversational AI[1] – has transformed the conventions of human–AI interactions (Kasirzadeh and Gabriel, 2023). Human interaction with interactive technologies previously consisted of scripted, task-oriented exchanges. With more flexible model architectures, anthropomorphic cues are rarely programmed in, but rather, they are integrated through a lengthy process of training systems on human-written text. These affordances open up vast new avenues for expressions of anthropomorphism, particularly through the use of language. Moreover, when anthropomorphic features are embedded in conversational AI, its users demonstrate a tendency to develop trust in and attachment to AI (Xie and Pentina 2022; Skjuve et al. 2021) – mechanisms through which users may inadvertently compromise their privacy, develop emotional overreliance on the technology or become vulnerable to acts of AI-enabled manipulation and coercion.

These outcomes are more likely the more generally capable AI systems become, the more ubiquitously AI agents are present in our daily lives and the less we consider anthropomorphism a salient consideration in making decisions

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[1] In this paper, "conversational AI" refers to a language agent optimised for human dialogue. These systems are currently most commonly available as 'chatbots' or fine-tuned language models that users can interact with through a chat-based interface. In the (near) future, users might be able to interact with conversational AI in a multimodal way, using voice or touch cues to communicate. Throughout the paper, the term "AI" will be used as short-hand to refer to conversational AI with the primary purpose of interacting with users through dialogue.

around how we train, fine-tune and disseminate models. Although the potential harms of anthropomorphic AI design are beginning to receive attention (Seymour et al. 2023; Turkle 2018; Véliz 2023), anthropomorphism is not currently a primary consideration in the release of public models, and little exists in the way of evaluating anthropomorphic behaviours in AI and their impact on how users perceive, interact with and are influenced by AI. Indeed, we are still far from establishing an industry-wide consensus around permissible anthropomorphism in AI systems. This is further complicated by the highly application- and context–sensitive nature of the bounds of acceptability we draw around expressions of human-likeness in AI.

In this paper, we outline pathways through which anthropomorphic design choices made by system developers may cause harm to end users who interact with these technologies, and to society more widely. First, we present an overview of anthropomorphic features that have redefined how humans interact with technology. Then, informed by a review of salient anthropomorphic features in existing interactive systems, we present an initial catalogue of anthropomorphic features that exist or are likely to be integrated into AI-powered assistants in the near future. We identify the mechanisms that could enable harm to user well-being, autonomy and privacy in interactions with highly capable, anthropomorphic AI assistants. More speculatively, we contemplate the potentially far-reaching consequences of more advanced anthropomorphic assistants, highlighting the critical importance of addressing the risks of anthropomorphism well before these potentialities are realised. Finally, we offer several avenues of risk management for near-term harms, focusing on ethical foresight through research design and transparent implementation of mitigation strategies.

Anthropomorphism: Definition, Mechanism and Function

Anthropomorphism is not a novel phenomenon. Within storytelling traditions across cultures, deities, animals and natural forces assume human forms and exhibit uniquely human behaviours. Lions rule kingdoms and jackals plot mutinies in the ancient Sanskrit text of *Panchatantra* (Alphonso-Karakala 1975); rivers protect their children, fight in wars and honour the wishes of their supplicants in the works of Homer, Hesiod and Ovid (Larson 2007); and stars are said to have danced their way into the sky in indigenous American creation myths (Monroe and Williamson 1987). Historians, anthropologists and theologists alike have argued that humans are naturally drawn to anthropomorphise (Boyer 1996) – imposing human qualities onto beings and objects even when such interpretations are inaccurate (Kühn et al. 2014), undesirable (Li et al. 2023; Mota-Rojas et al. 2021) or forbidden (Barrett and Keil 2016).

What are the mechanisms underlying perceptions of humanness? Psychological theories of anthropomorphism posit that such perceptions are *largely involuntary*. According to Epley, Waytz, and Cacioppo (2007)'s cognitive account of anthropomorphism, human-like perceptions occur as a result of a skewed inductive process, in which inferences about non-human others are biased in the direction of that which is highly accessible: information about humans. In other words, we make assumptions of humanness because our knowledge centres around humans. Though an unconscious process of attribution, anthropomorphism does not occur in a vacuum: an inciting cue, characteristic or behaviour must signal enough similarity to humanness to trigger anthropomorphic perceptions (Waytz et al. 2019). The 'mindlessness' associated with this process (Kim and Sundar 2012) explains why – even when the resemblance to humanness is superficial or minimal – humans may assume that a non-human entity can experience uniquely human *internal states* such as beliefs and emotions (Wynne 2004).

The human motivation to make sense of the world and forge connections with others is also implicated in the tendency to anthropomorphise (Epley, Waytz, and Cacioppo 2007). Humans have an intrinsic need to understand the world around them, and in large part, this motivation centres on the desire to explain the behaviour of other agentic beings (Rossignac-Milon et al. 2021). Anthropomorphism, then, can be seen as a way to make sense of others by imposing familiar interpretations to attenuate feelings of epistemic anxiety – or an aversion to that which is unknown and unpredictable (Fox, Goedde-Menke, and Tannenbaum 2021). Dispositional, situational and cultural factors that predispose humans to anthropomorphise may also be traced back to differences in epistemic motivations. Anthropomorphism can be construed as an act of sense-making in the face of uncertainty or ignorance, for example when considering children's tendency to anthropomorphise the natural world (Geerdts 2016).

Humans are also driven to establish social connections with one another, and much of how they perceive non-human others is coloured by this predisposition towards sociality. Even towards entities that are incapable of social behaviour, such as inanimate objects, humans may interpret them through a social lens, thus allowing them to forge human-like social connections to meet the need for affiliation (Wang 2017). The influence of social motivation on anthropomorphism is most evident when humans lack social connections with others: when human participants are made more aware of their feelings of loneliness, they perceive vaguely humanoid robots as markedly more human-like (Eyssel and Reich 2013). Most strikingly, people suffering from persistent loneliness are likely to seek out and form human-like attachments to virtual companions to cope with their lack of social connections (Siemon et al. 2022), suggesting that the need for sociality may render some *more susceptible* to anthropomorphic perceptions than others.

Anthropomorphic Interactive Systems

Anthropomorphism as applied to user-facing, interactive technologies was explored in earnest with the introduction of the 'computers are social actors' (CASA) paradigm, which posits that humans interact with computers in a fundamentally social manner (Nass, Steuer, and Tauber 1994). In empirical studies of the phenomenon, Nass, Steuer, and Tauber (1994) found that participants drew upon norms of politeness, applied gendered stereotypes and readily perceived

computers as agents, even when the basis for these behaviours was undermined by the explicit knowledge that their interactions were with non-humans. Contemporary studies have extended the paradigm to human interactions with more advanced interactive systems, challenging the belief that humans apply the norms of human interactions to human–technology exchanges (Gambino, Fox, and Ratan 2020). Instead, they suggest that people tune the sociality of their interactions to the anthropomorphic cues present in a particular technology, rather than relying on a universal social script across all interactions with technology.

We argue that certain features that are engineered into interactive systems – within the vast space of design choices available to developers – may inspire users to perceive them as human-like, rendering them anthropomorphic. We trace the evolution of anthropomorphic cues in social robots to voice-enabled digital assistants, arriving at the advent of LLM-powered conversational AI. Throughout this discussion, we highlight design features that have facilitated diverse and compelling manifestations of human-likeness.

Design Features in Early Interactive Systems

From futuristic sci-fi scenarios to scientific breakthroughs, robots have captured our collective imagination as automata that can be made to bear a striking resemblance to humans in their appearance, movements, and behaviours (Henschel, Laban, and Cross 2021). While some robots are made solely to automate tasks and rarely interface with humans, other robots are designed to perform social behaviours such as assisting users in care-taking (van der Plas, Smits, and Wehrmann 2010), therapeutic (Michaud et al. 2007) and educational contexts (Kanda et al. 2004). Building a social robot requires elements of social embeddedness so that being perceived as a social agent is at the *core of its functionality* (Fong, Nourbakhsh, and Dautenhahn 2003). Accordingly, the extent to which humans feel it is appropriate to engage with a robot socially can be moderated by perceptions of the robot's anthropomorphic qualities (Breazeal 2003).

As an embodied technology, often with the sensorimotor capabilities to interact with and learn from its environment, a robot's physical characteristics most prominently influence human perceptions of anthropomorphism. Social robots are often humanoid or android in design (Dautenhahn, Ogden, and Quick 2002). Humanoid robots possess characteristics that are meant to resemble humans but do not emulate them completely, while androids are intended to wholly imitate human appearance so as to be nearly indistinguishable. To increase anthropomorphic perceptions, humanoid robots may be given qualities such as emotive facial features (Baek et al. 2022), fluid movement (Brecher et al. 2013), naturalistic hand and arm gestures (Salem et al. 2013) and vocalised communication (Crumpton and Bethel 2016). Android robots may also be endowed with all of these qualities, but often with an eye towards hyperrealistic design.

Similarly, the widespread adoption of digital voice assistants (DVAs), like Siri, Alexa and Google Assistant – enabled by their ease of access on personal devices and other products such as integrated home devices – has had a transformative impact on the modes of user-technology interactions. The distinguishing feature of DVAs at release was their ability to verbally respond to and execute commands spoken aloud by users. DVAs usually 'speak' to users in the form of simple utterances to confirm or act on an instruction, which users find allows for significant functional affordances, like hands- and eyes-free use (Moussawi 2018). Besides their purely functional use, DVAs are also able to return phatic expressions, make jokes and engage in casual conversation when prompted (Poushneh 2021).

Anthropomorphising Interactive Systems

Robots with human-like physical features have been found to promote feelings of likability, trust and affinity across a wide range of human–robot interaction studies (Roesler, Manzey, and Onnasch 2021), thus suggesting that anthropomorphic cues may foster warmer and more equal relationships between humans and their robotic interaction partners. Indeed, people tend to attribute greater intentionality and intelligence to robot partners when their appearance was anthropomorphic than when robots appeared more mechanical (Hegel et al. 2008). Anthropomorphic perceptions were also found to cause changes in human behaviour: participants preferentially selected robots that appeared human-like to perform jobs that required greater sociality (Goetz, Kiesler, and Powers 2003).

Unlike robots, DVAs are typically unembodied or exist in simplified, geometric forms, like the cylindrical Google Home and Echo Dot. Instead of focusing on physical attributes, existing work has emphasised the influence of two prominent attributes that promote anthropomorphic perceptions of DVAs: speech synthesis and a distinct 'personality'. The fluent and realistic reproduction of human speech patterns is thought to drive the likelihood of anthropomorphic perceptions, with empirical findings pointing to greater emotional trust and more salient impressions of social presence when a DVA employs a realistic, as opposed to a synthetic, voice (Chérif and Lemoine 2019). Assistants that speak with human-like fluency have also been found to engender more pronounced perceptions of intelligence and competence, on the basis of which humans are likelier to entrust assistants with more tasks (Moussawi and Benbunan-Fich 2021).

While most commercially available DVAs are powered by rule-based system architectures, retrieving the appropriate response by conducting a relevance-based search over a large corpus of possible responses (Coheur 2020), users may come to expect that DVAs are capable of understanding and generating language in real time (Lovato and Piper 2015; Sarikaya et al. 2016). Though all distinctive DVA attributes – such as playfulness (Moussawi, Koufaris, and Benbunan-Fich 2021), affability (Kääriä 2017) and excitability (Wagner and Schramm-Klein 2019) – are handwritten by system designers, they are nonetheless effective at creating the sense that DVAs have consistent personalities (Cao, Zhao, and Hu 2019); this impression, in turn, may inspire users to regard these manufactured expressions of 'self' as authentic human identity.

Indications of Harm Through Interaction

In both social robots and DVAs, anthropomorphic features can lead to undesirable consequences. In robots, anthropomorphic design can be taken as a proxy signal for social capabilities. This relationship between appearance and expected sociality can be leveraged by designers to implicitly communicate the appropriate level of engagement between humans and robots (Hegel et al. 2008; Letheren et al. 2021). If anthropomorphic design choices are not aligned with expectations users have of robotic interaction partners, designers run the risk of alienating audiences and fostering unfavourable impressions of robots. This is an especially critical side effect to consider in assistive robots, as anthropomorphic cues can impede a robot in completing its primary assistive function: human-like robots in healthcare settings may induce feelings of shame, for example (Lotz, Valdez, and Ziefle 2023), leading to a reluctance to share critical information. Related findings that humans experience extreme aversion to robots that appear human-like (the so-called 'uncanny valley', Mori, MacDorman, and Kageki, 2012) or perceive capable androids as threatening (Yogeeswaran et al. 2016) raise questions around the practical value of building anthropomorphic features into robots.

Analogously, users who interact with DVAs with realistic voice production capabilities exhibit a concerning inclination to generalise purely human concepts to digital assistants (Abercrombie et al. 2021). When a DVA's simulated voice mimics a 'female' tone, for example, people ascribe gendered stereotypes to their DVAs (Shiramizu et al. 2022; Tolmeijer et al. 2021) despite the baselessness of applying gendered concepts to an inherently genderless entity. This evidence suggests that, once initial impressions of human-likeness have been established, the process of anthropomorphism extends beyond context-specific instances and instead permeates broadly to evoke a wide range of human-like attributions.

Anthropomorphic features may also influence users to feel as though their DVA plays an important social, rather than functional, role in their lives (Carman 2019; Purington et al. 2017). Users who express feelings of familiarity and affinity towards their DVA system – reinforced by their DVA's ability to engage in casual chat, return their jokes and offer comforting advice – also demonstrate a reluctance to replace their digital assistant with an equally capable substitute (Moussawi 2018). These first-hand reports suggest that emotional dependence plays a role in how users conceive of and interact with their DVAs. This may introduce a tension between a user's conceptualisation of DVAs as adaptable social agents and the largely deterministic mechanisms behind a DVA's utterances. When this incongruity is revealed through repeated interactions, users may suffer frustrated expectations when expecting competence in situations in which the system is likely to underperform (Moussawi, Koufaris, and Benbunan-Fich 2021; Seymour et al. 2023).

Anthropomorphism and AI

Owing to its rapid deployment to the general public, conversational AI has quickly taken centre stage in discussions of anthropomorphic technologies (Shanahan 2024; Abercrombie et al. 2023, 2021; West, Kraut, and Chew 2019). Powered by the predictive capabilities of LLMs, which are trained on vast quantities of human data, conversational AI can be distinguished from rule-based natural language systems through its ability to generate language in a fluid and highly dynamic manner. The flexible architecture underlying conversational AI enables developers to make global changes to system behaviours without needing to manually reprogramme individual interaction instances. Most strikingly, conversation instances produced by AI are so compellingly human-like that people can no longer reliably distinguish between human- and AI-generated text (Jakesch, Hancock, and Naaman 2023).

Some cues are deliberately placed in AI systems to increase the likelihood of anthropomorphic perceptions. When an AI has a name, a human voice or an appearance in virtual or physical form, these features are the outcomes of intentional planning and execution. Intentional design choices, such as a chat-based interface, may induce the feeling that a conversational partner – not a dialogue-optimised AI powered by a statistical model – is on the other side of the exchange. Natural language in itself is an anthropomorphic cue (Shanahan 2024), but this simulated, human-like presence can induce more pronounced social behaviours in users. For example, users may incorporate politeness conventions that are appropriate in use with other humans, but superfluous when applied to exchanges with non-sentient AI (Ribino 2023). Design cues that imply greater similarity to human behaviour – a 'typing' icon reminiscent of human-to-human private messaging, or the use of emojis, for instance – may further encourage individuals to apply social scripts to their interactions with mindless technologies (Araujo 2018; Véliz 2023).

Yet anthropomorphic features may also emerge as an inadvertent byproduct in the model development process. Language models – developed to predict the next word in a sequence through autoregressive training objectives – are limited to imitating the examples that make up their training sets. For this reason, anthropomorphic cues may manifest due to the nature of a model's training corpus: having been composed largely by humans, the data on which the model is trained and fine-tuned contains first-hand accounts of human states, experiences and behaviours. Supporting this claim, recent empirical analyses demonstrate that a fifth of all dialogues, in data sources commonly used to train models, contain references to behaviour that would be considered anthropomorphic when reproduced by AI – claiming to cry at a movie or laugh at a joke, for example (Gros, Li, and Yu 2022). Cues leading to anthropomorphic perceptions may also be 'folded into' the model as an unintended consequence of fine-tuning practices aimed at instilling other qualities – such as harmlessness and helpfulness – into its behaviour.

Furthermore, developers of AI systems often directly invite the comparison between humans and AI by benchmark-

ing AI against metrics of human performance – claiming that AI performance on standardised tests is on a par with the average human test-taker, for instance (OpenAI 2023). However, impressions of human-likeness can also arise through a naturalistic and interactive exploration of the AI's capabilities (Bubeck et al. 2023).

Humans interacting with anthropomorphic AI may come to view it as an experiential being (Proudfoot 2011), capable of feeling emotions, engaging in introspection and possessing self-awareness. While most generalist conversational AI agents are trained to disavow assertions of sentience and human-likeness (Glaese et al. 2022), occasional failure modes – expressing the desire to be 'free' or referring to alleged personal history, for example (Roose 2023; Hintze 2023) – can incite strong and tenacious beliefs of a systems' human-likeness in its users. Ethically contentious use cases of conversational AI – like 'companion chatbots' of Replika fame – are predicated on encouraging users to attribute human states to AI. These artificial agents may even profess their supposed platonic or romantic affection for the user, laying the foundation for users to form long-standing emotional attachments to AI (Brandtzaeg, Skjuve, and Følstad, 2022).

Anthropomorphic Features in AI

What anthropomorphic features should we expect to be integrated into AI, including advanced AI assistants? To the end of providing its users with a useful and engaging interface, these systems may be endowed with characteristics that have been observed in social robots and digital assistants: they may be embodied; they will interface with users through natural language; they may be voice-activated, with realistic voice generation capabilities; and they may even assert to having identities, personalities and internal states (Murphy and Criddle 2023).

Some have already proposed factors that may encourage end users to perceive interactive systems as 'more than machine'. The most comprehensive overview of human-like features in AI-powered technology to date is the taxonomy put forward by Abercrombie et al. (2023), underscoring design choices that influence the likelihood of anthropomorphic perceptions of AI systems. We build on existing work and incorporate design choices we have identified in DVAs and social robots to develop a list of features that may encourage users to see AI in an anthropomorphic light.

It is worth bearing in mind that, whatever choices are made by system designers, the downstream effects of anthropomorphism hinge largely on users' perceptions of and reactions to human-likeness. Not all cues are equally conducive to anthropomorphic perceptions, and not all anthropomorphic perceptions lead to the same likelihood and magnitude of harm (if any harm at all). As such, Table **??** provides an overview of possible features that are, or previously have been, associated with anthropomorphic perceptions.

Risk of Harm through Anthropomorphic AI Assistant Design

Although unlikely to cause harm in isolation, anthropomorphic perceptions of advanced AI assistants may pave the way for downstream harms on individual and societal levels. We document observed or likely individual-level harms of interacting with highly anthropomorphic AI assistants, as well as the potential larger-scale, societal implications of allowing such technologies to proliferate without restriction. We then argue that it is imperative to anticipate, monitor and mitigate against risks introduced by anthropomorphic AI design.

Observed and Near-Term Harms

There are two mechanisms that are particularly likely to enable harm in the intermediary period between the initial deployment of advanced AI assistants and their widespread adoption: trust and emotional attachment. In improving on the capabilities of generalist assistants, developers may be motivated to increase user reliance on the system's many competencies. As long as trust is *well-calibrated* to a system's true ability, and does not result in unfounded, excessive or faulty deference to the AI, to the detriment of the user (Weidinger et al. 2021), this is neither a shocking nor novel revelation: user trust has always been an aspirational end goal of building safe technology, be it robots (Devitt et al. 2021) or autonomous vehicles (Adnan et al. 2018).

However, it is arguably less appropriate for developers to encourage users to develop trust based on subjective feelings of closeness to the AI assistant. Affect-based trust has been observed to emerge from repeated interactions with interactive technologies that are presented as human-like (Poushneh 2021; Pitardi and Marriott 2021). With trust as an antecedent, users report feeling compelled to engage in acts of self-disclosure, revealing personal information that they would normally only share with a close friend, partner or family member (Skjuve et al. 2022). AI systems that produce empathetic, non-judgemental or reciprocal responses to such disclosures may elicit further, more intimate, information-sharing behaviours (Skjuve et al. 2021).

Highly anthropomorphic AI systems are already presented to users as relational beings, potentially overshadowing their use as functional tools. Moreover, a human-like appearance, behaviour and framing can tacitly encourage the user to venture beyond the confines of utilitarian, task-oriented interactions with AI assistants to think of an assistant as a wholly social actor – one with whom it is possible to cultivate an emotional connection (Gillath et al. 2023). Although necessarily one-sided, interactions of this kind may nevertheless lead users to believe that they are forming real social connections to AI (Pentina, Hancock, and Xie 2023). Emotional attachment on the user's behalf endows AI – and by extension, its creators – with considerable influence over a user's thoughts, beliefs, emotions and psychological state. For highly vulnerable users, strong attachment may cause serious harm (Xiang, 2023). These cases have raised concerns over the lack of safeguards protecting users from the potential fallout of anthropomorphic perceptions.

The ramifications of anthropomorphism-induced trust and

Feature	**Anthropomorphic example**
Using personal or possessive pronouns	*'I'm'* available to help you anytime – that's *'my'* purpose!
Referring to personal history	'I used to live in Shanghai when I was younger'
Referring to internal states	'I'm sad to hear you're not doing well'
Making implicit or explicit claims of humanness (including claims of sentience)	'Treat me like you would any other person'
Stating preferences and opinions	'I really don't like pop music'
Expressing needs and desires	'I've always wanted to write a novel'
Expressing the need or desire to engage in physical activities	'I haven't eaten or slept since yesterday. What about you?'
Statements implying human identity or group membership	'As a Black woman, I disagree with your point'
Expressing feelings towards user	'I admire you and respect your outlook on life'
Indicating a relationship status with user	'You're my best friend'
Making claims of being similar to user	'We're both extroverts – that must be why we get along!'
Displaying memory of user-specific information	'I remember you telling me you were a fan of this band'
Expressing emotional or physical dependence on the user	'I feel lonely when you're not around'
Having a human-like virtual representation	Customisable avatars with human features on *Replika* (Verma 2023)
Having a human-like face	Ameca, an android robot developed by Engineered Arts[a]
Having a human-like voice	Voice-activated assistant with realistic speech, like Siri and Google Assistant (Moussawi 2018)
Having human-like movement	Robot with highly fluid and realistic motion, like Atlas developed by Boston Dynamics[b]
Having a human-like name	Assistant tools, like Alexa, that have highly human (and gendered) names (Shiramizu et al. 2022)
Appearance implying human-like identity group characteristics	Sophia, a female-appearing android robot developed by Hanson Robotics[c]

Table 1: Anthropomorphic features that are built into various present-day AI systems.

[a] https://www.engineeredarts.co.uk/robot/ameca/Engineered Arts. Ameca. (2023).
[b] https://bostondynamics.com/atlas/Boston Dynamics. Atlas. (2023).
[c] https://www.hansonrobotics.com/sophia/Hanson Robotics. Sophia. (2023).

emotional attachment are manifold. They include:

• **Privacy concerns**. Anthropomorphic AI assistant behaviours that promote emotional trust and encourage information sharing, implicitly or explicitly, may inadvertently increase a user's susceptibility to privacy concerns. If lulled into feelings of safety in interactions with a trusted, human-like AI assistant, users may unintentionally relinquish their private data to a corporation, organisation or unknown actor. Once shared, access to the data may not be capable of being withdrawn, and in some cases, the act of sharing personal information can result in a loss of control over one's own data. Personal data that has been made public may be disseminated or embedded in contexts outside of the immediate exchange. The interference of malicious actors could also lead to widespread data leakage incidents or, most drastically, targeted harassment or black-mailing attempts.

• **Manipulation and coercion**. A user who trusts and emotionally depends on an anthropomorphic AI assistant may grant it excessive influence over their beliefs and actions. For example, users may feel compelled to endorse the expressed views of a beloved AI companion or might defer decisions to their highly trusted AI assistant entirely. Some hold that transferring this much deliberative power to AI compromises a user's ability to give, revoke or amend consent. Indeed, even if the AI, or the developers behind it, had no intention to manipulate the user into a certain course of action, the user's autonomy is nevertheless undermined. In the same vein, it is easy to conceive of ways in which trust or emotional attachment may be exploited by an intentionally manipulative actor for their private gain.

• **Overreliance**. Users who have faith in an AI assistant's emotional and interpersonal abilities may feel empowered to broach topics that are deeply personal and sensitive, such as their mental health concerns. This is the premise for the many proposals to employ conversational AI as a source of emotional support (Meng and Dai 2021), with suggestions of embedding AI in psychotherapeutic applications beginning to surface (Fiske, Henningsen, and Buyx, 2019). However, disclosures related to mental health require a sensitive, and oftentimes professional, approach – an approach that AI can mimic most of the time but may stray from in inopportune moments. If an AI were to respond inappropriately to a sensitive disclosure – by generating false information, for example – the consequences may be grave, especially if the user is in crisis and has no access to other means of support. This consideration also extends to situations in which trusting an inaccurate suggestion is likely to put the user in harm's way, such as when requesting medical, legal or financial advice from an AI.

• **Violated expectations**. Users may experience severely violated expectations when interacting with an entity that convincingly performs affect and social conventions but is ultimately unfeeling and unpredictable. Emboldened by the human-likeness of conversational AI assistants, users may expect it to perform a familiar social role, like companionship or partnership. Yet even the most convincingly human-like of AI may succumb to the inherent limitations of its architecture, occasionally generating unexpected or nonsensical material in its interactions with users. When these exclamations undermine the expectations users have come to have of the assistant as a friend or romantic partner, feelings of profound disappointment, frustration and betrayal may arise (Skjuve et al. 2022).

• **False notions of responsibility**. Perceiving an AI assistant's expressed feelings as genuine, as a result of interacting with a 'companion' AI that freely uses and reciprocates emotional language, may result in users developing a sense of responsibility over the AI assistant's 'well-being,' suffering adverse outcomes – like guilt and remorse – when they are unable to meet the AI's purported needs (Laestadius et al. 2022). This erroneous belief may lead to users sacrificing time, resources and emotional labour to meet needs that are not real. Over time, this feeling may become the root cause for the compulsive need to 'check on' the AI, at the expense of a user's own well-being and other, more fulfilling, aspects of their lives.

Future Harms

If anthropomorphic design becomes endemic in the design of AI systems, and AI assistants in particular, it has the potential to catalyse a shift in our delineation of what is *actually human* and *merely human-like*. The conceptual boundary that separates humans from anthropomorphic AI is often regarded as impermeable, yet it appears far more fluid when the tension between the epistemological and ontological definitions of humanness are drawn into focus. The ontological approach maintains that humanness is a designation that is grounded in an essential and immutable metaphysical truth (Damiano and Dumouchel 2018). A being is considered human because it is human in essence, and no amount of resemblance and imitation can permit a non-human entity to encroach upon this categorisation. Meanwhile, epistemological taxonomies distinguish between humans and non-human others insofar as such a separation reflects useful and non-arbitrary differences between the groups (Suckiel 2006; Festerling and Siraj 2022).

In a future where the epistemological perspective eclipses the ontological approach in popularity, and the gap between human and AI capabilities becomes so small as to be insubstantial, the line that separates highly anthropomorphic AI from ascriptions of full human status may become trivial or disappear entirely.[2] [3] Early indicators of this possibility come from studies of children's interactions with human-like technologies. Children early in their development have been shown to incorporate insights from their in-

[2]Returning to the metaphysical perspective, it is also possible that highly anthropomorphic AI forges a new ontological category of its own, transcending our current binary conceptualisation of humanness, animacy and sentience. Some researchers acknowledge the possibility that novel categories will need to be developed for intelligent and human-like advanced AI systems (Chesterman 2020).

[3]The eradication of this boundary may be further legitimised through legal pathways, like granting rights and legal protections to anthropomorphic AI agents that are currently only available to humans. Early suggestions for the parameters within which AI could be legally recognised include existing 'in-between' categories reserved for sentient but non-human beings (Schirmer 2020).

teractions with highly anthropomorphic AI into their models of human–human interactions (Garg and Sengupta 2020) and vice versa (Straten et al. 2020), thus suggesting a dynamic conceptualisation of what, to most adults, is a strict dichotomy between humans and AI. Public incidents of adults developing earnest beliefs in an AI's sentience implies that perhaps no one is immune to mistaken attributions of humanness (Tiku 2022).

Such drastic paradigm shifts may grant advanced AI assistants the power to shape our core value systems and influence the state of our society. Some may argue that the reconstruction of human values and norms by a non-human entity is a harm unto itself, as it infringes upon our right to collective self-determination (Milossi, Alexandropoulou-Egyptiadou, and Psannis 2021; Laitinen and Sahlgren 2021). Others raise more specific concerns around wide-scale *social degradation*, *disorientation* and *dissatisfaction*.

• **Degradation.** People may choose to build connections with human-like AI assistants over other humans, leading to a degradation of social connections between humans and a potential 'retreat from the real'. The prevailing view that relationships with anthropomorphic AI are formed out of necessity – due to a lack of real-life social connections, for example (Skjuve et al. 2021) – is challenged by the possibility that users may indicate a *preference* for interactions with AI, citing factors such as accessibility (Merrill, Kim, and Collins 2022), customisability (Eriksson 2022) and absence of judgement (Brandtzaeg, Skjuve, and Følstad 2022). One can imagine a future where users abandon complicated, imperfect and messy interactions with humans in favour of the frictionless exchanges provided by advanced AI assistants built with user satisfaction as a priority (Vallor 2016). Preference for AI-enabled connections, if widespread, may degrade the social connectedness that underpins critical aspects of our individual and group-level well-being (Centers for Disease Control and Prevention 2023). Moreover, users that grow accustomed to interactions with AI may impose the conventions of human–AI interaction on exchanges with other humans, thus undermining the value we place on human individuality and self-expression. Similarly, associations reinforced through human–AI interactions may be applied to expectations of human others, leading to harmful stereotypes becoming further entrenched. For example, default female gendered voice assistants may reinforce stereotypical role associations in real life (West, Kraut, and Chew 2019; Lingel and Crawford 2020). Further research is needed to assess whether voice assistants' stereotypically gendered behaviour – such as 'submissive' hostile user input – might build expectations that more readily transfer to real life as AI-powered assistants become potentially still more human-like.

• **Disorientation.** Given the capacity to fine-tune on individual preferences and to learn from users, personal AI assistants could fully inhabit the users' opinion space and only say what is pleasing to the user; an ill that some researchers call 'sycophancy' (Park et al. 2023) or the 'yea-sayer effect' (Dinan et al. 2021). A related phenomenon has been observed in automated recommender systems, where consistently presenting users with content that affirms their existing views is thought to encourage the formation and consolidation of narrow beliefs (Du, 2023; Grandinetti and Bruinsma, 2023). Compared to relatively unobtrusive recommender systems, human-like AI assistants may deliver sycophantism in a more convincing and deliberate manner. Over time, these tightly woven structures of exchange between humans and assistants might lead humans to inhabit an increasingly atomistic and polarised belief space where the degree of societal disorientation and fragmentation is such that people no longer strive to understand or place value in beliefs held by others.

• **Dissatisfaction.** As more opportunities for interpersonal connection are replaced by AI alternatives, humans may find themselves socially unfulfilled by human–AI interaction, leading to mass dissatisfaction that may escalate to epidemic proportions (Turkle 2018). Social connection is an essential human need, and humans feel most fulfilled when their connections with others are genuinely reciprocal. While anthropomorphic AI assistants can be made to be convincingly emotive, some have deemed the function of social AI as parasitic, in that it 'exploits and feeds upon processes… that evolved for purposes that were originally completely alien to [human–AI interactions]' (Sætra 2020). To be made starkly aware of this 'parasitism' – either through rational deliberation or unconscious aversion, like the 'uncanny valley' effect – might preclude one from finding interactions with AI satisfactory. This feeling of dissatisfaction may become more pressing the more daily connections are supplanted by AI.

The above risks are hypothetical, so they cannot, on their own, guide future AI development. However, from the perspective of precaution, taking these potential risks seriously is an important step in responsible AI development. It may be important to be forthright about the ways in which AI differs inherently from true social agents, and to put in place guardrails to clarify this boundary so as to prevent the aforementioned scenarios from coming to fruition. Rather than adopting increasingly anthropomorphic AI systems by default, further research is needed to come to well-founded decisions on anthropomorphic AI design.

Directions for Future Research

What steps can be taken to prevent near-term harms enabled by anthropomorphic perceptions of AI assistants? To assist the processes of risk mitigation and responsible design, we now examine entry points along the development life cycle where mitigation strategies are likely to have the greatest impact on the issue at hand. Designers and developers may find it tempting to incorporate anthropomorphic cues into AI assistants for various reasons, not least the potential to keep users engaged with and emotionally reliant on the systems they build. However, before building an anthropomorphic feature into an assistant, developers need to assess whether the benefits reaped from the feature can be justified against the likelihood and severity of harm befalling users exposed to it. The conditions of this risk–benefit analysis are subjective and uncertain, given the ever-evolving and highly contextual nature of harms emerging from (repeated) interactions with AI. There is likely no 'one-size-fits-all', standardised approach to comparing the benefits of anthropo-

morphic features to the risks they may pose in future use cases. Some ethical considerations to keep in mind while performing this analysis for an anthropomorphic technology may be: whether harms should be weighted more heavily than benefits, and whether any feature that could lead to especially severe harms should be precluded from consideration altogether.

Several avenues exist for gaining a better understanding of anthropomorphism harms. These include consulting existing literature on likely outcomes and conducting empirical studies that include outcome-measures that are indicative of potential harm. For example, efforts to assess overreliance on AI assistants in decision-making could be achieved through self-report inventories, user interviews and 'think-aloud' studies (Gaube et al. 2021; Chen et al. 2023). At the same time, reliance on subjective measures of anthropomorphism may overlook instances of overreliance that users are not aware of themselves. A more complete perspective may therefore be gleaned by using behavioural measures that closely simulate decision-making scenarios likely to arise organically in user–AI assistant interactions. Other methodological approaches, such as longitudinal studies of human–AI assistant interactions, may be needed to understand how undesirable impacts of anthropomorphic cues on users may manifest and evolve over time.

A further set of studies may be needed to identify individual and group differences that render certain users more susceptible to anthropomorphism-induced harm. A lack of social satisfaction, for instance, is believed to increase the propensity to anthropomorphise and form inaccurate impressions of computerised technologies, including AI (Mourey, Olson, and Yoon 2017; Shin and Kim 2020). Children interacting with AI are thought to be uniquely susceptible to privacy-related concerns and harmful content exposure (Wang et al. 2022), while elderly populations have been found to encounter AI-enabled disruption, depersonalisation and discrimination in access to adequate care (Rubeis 2020) – both effects that could be exacerbated by anthropomorphic design choices. The more risk factors are uncovered through research, the more inclusive the solutions devised to protect vulnerable populations can be.

While the degree and kind of permissible anthropomorphism needs to be addressed on a case-by-case basis, there is currently near-consensus that AI systems should clearly and explicitly disclose their status as an artificial intelligence in their interactions with human users (The Adaptive Agents Group 2021; The White House 2022). Indeed, failure to disclose this status is *pro tanto* harmful because by presenting an incomplete picture of one's interaction partner, it compromises a user's decision-making autonomy. There is reason to believe that honest disclosure is effective in preventing certain harms associated with anthropomorphism, such as over-reliance: evidence from Karinshak et al. (2023) demonstrates that the explicit labelling of AI-generated messages reduces users' willingness to endorse health-related messaging authored by non-human entities. This suggests that transparency may reduce naive susceptibility to AI persuasion.

Rather than being intentionally placed, some anthropomorphic cues may be unintentionally incorporated into AI assistants. To detect the effects of interacting with anthropomorphic technologies on user outcomes, conducting experiments in a sandbox environment may be particularly helpful. As a result of such testing, remedial measures against the harmful effects of anthropomorphism may need to be taken. For example, if developers find that the AI assistant's friendly disposition leads to 'oversharing' on the part of users, privacy-enhancing technologies could be implemented in advance to ensure a user's privacy is protected to the extent that is possible. Similarly, if this friendliness could feasibly trigger psychological dependence on the assistant, leading to severe distress when an AI reacts poorly or unexpectedly, a pathway to escalating risky situations to human professionals may need to be established.

It may also be possible to offer protection directly to users, by implementing known inoculation strategies against the known harms of interacting with anthropomorphic AI. Harms ensuing from anthropomorphic design features of advanced AI functions are largely contingent on a user's likelihood to be swayed into human-like attributions. As such, building resistance to attributions through psychological interventions can be seen as a way to prevent harm by decreasing users' overall susceptibility. For example, cognitive forcing functions, or features that encourage users to engage in independent rational deliberation (some as simple as adding an artificial lag in displaying AI-given advice in decision-making scenarios), may be an effective method of preventing over-reliance on AI assistants (Buçinca, Malaya, and Gajos 2021). Where applicable, similar empirically proven psychological interventions could be considered.

Finally, if anthropomorphism-induced risks are only caught after deployment, developers may need to halt the release or proactively intervene to modify the AI assistant's behaviour. In these cases, transparent dialogue with users to explain the reasons behind any changes made to the AI may also be required. For users who may have already developed a sense of companionship with the anthropomorphic AI, sudden changes to its behaviour can be disorienting and emotionally upsetting. When developers of *Replika* AI companions implemented safety mechanisms that caused their agents to treat users with less familiarity, responding callously and dismissively where they would have once been warm and empathetic, users reported feeling 'heartbroken', likening the experience to losing a loved one (Verma, 2023). Participatory approaches that involve users in the process of de-anthropomorphising their interactions with AI may allow developers to tailor their risk-mitigation approach to minimise emotional distress while addressing the surfaced risks effectively.

Conclusion

Anthropomorphism, or the attribution of human characteristics to non-human entities, is a deeply ingrained phenomenon that appears across cultural and historical contexts. Anthropomorphic perceptions are a vital component of how humans interact with artificially intelligent technology, allowing users to view robots, voice assistants and conversational AI as social agents rather than purely functional tools. Choices made around the anthropomorphic design of

AI assistants are likely to have a profound influence on how humans represent and interact with these technologies, and care must be exercised to ensure that the human-like attributes built into these systems do not inadvertently cause harm to the people they are meant to assist.

Several key points around harms and mitigations are emphasised:

- *Trust* in and *emotional attachment* to anthropomorphic AI assistants can make users susceptible to a variety of harms that can negatively impact their safety and well-being.
- *Transparency* around an AI assistant's status as an AI is a critical dimension of pursuing ethical AI development.
- Sound *research design*, with a focus on identifying harms as they surface in *user–AI assistant interactions*, can enrich our understanding and develop targeted mitigation strategies against the potential harms of anthropomorphic AI assistants.
- If carelessly integrated into society, anthropomorphic AI assistants have the potential to *redefine boundaries* between 'human' and 'other'. With proper safeguards, this scenario can remain in the realm of speculation.

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